Eclipses by Appointment: 470 Eclipsing Systems, Catalogue Corrections and a Measured Null Planet Search from Public TESS and Gaia DR3 Data

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The author's role and the use of AI systems are set out in Section 6, Contributions and AI use. Per-object analyses, available light curves and supporting measurements are in the Supplement and repository; catalogue corrections, probable blends and other objects are also documented in the earlier working draft (Appendix B).

AAVSO keywords = Eclipsing Binaries; Catalogues, Databases, Surveys; Period Analysis, Period Change; Extrasolar Planets; Variable Stars (individual), Observing Targets: HIP 1060, HD 170829, V1192 Cas, TYC 4590-364-1

ADS keywords = binaries: eclipsing; binaries: spectroscopic; catalogs; planetary systems; stars: variables: general; techniques: photometric

ABSTRACT

We identify 470 eclipsing systems in public TESS photometry: 439 confirmed and 31 probable. All 470 meet the specified seven-group catalogue standard. Fourteen further signals remain plausible and are excluded. One hundred seven systems come from a method we call eclipses by appointment: a Gaia DR3 spectroscopic orbit with a period of 20–60 d predicts conjunction windows in which TESS can reveal eclipses that a blind period search misses. Among 3,749 stars with G < 11 the method found about 1.7 new eclipsing binaries per hundred stars; a G 11–11.5 subsample contributes 42 confirmed systems. Gaia-orbit period folds contribute 355 systems, the nearby-M-dwarf-selected search contributes four (including neighbouring hosts), and near-edge-on Gaia astrometric orbits contribute four. The period-fold sample includes fainter spectroscopic binaries and two probable cases from a near-orbital-period pilot. An adaptive exploratory conjunction pilot adds two confirmed systems while retaining a known recovery separately; its selection and stopping do not support an unbiased yield estimate. The analyses combine raw-flux measurements, timing and coverage tests, Gaia covariance, alternative reductions and source attribution, with physical classification kept separate from scoped catalogue status. Twenty Gaia DR3 orbit solutions are contradicted under the paper's stated comparison rule; nine eccentric binaries are identified as new heartbeat stars, one also eclipsing, and a tenth is a candidate; eleven existing catalogue entries are corrected. Harvard plates from 1890–1990 support the eclipses of 17 of 31 tested systems, ASAS-SN photometry supports 20 of 55, and ASAS-3 supports 10 of 34. A transit search of about 3,100 nearby M dwarfs found no new planet where about 1.2 detections were expected under the stated assumptions. We recover both TIC 4206066 candidates and identify TOI-5394.01 as an M-dwarf occultation. Per-object evidence, ephemerides where constrained, status limitations and follow-up tests accompany the sample.

1. Introduction

Gaia DR3 published about 180,000 spectroscopic orbits (Gaia Collaboration, Arenou et al. 2023), each giving a period, an eccentricity, an argument of periastron and a time of periastron for a binary resolved only in velocity. TESS has photometered most of those stars in its full-frame images over the same years. Where the two overlap, an orbit seen within a degree or two of edge-on must eclipse, and TESS records the eclipse whenever a conjunction falls in an observed sector. For periods below about 20 d a period search at Gaia's period finds such eclipses easily. Between 20 and 60 d it does not: a 27-day sector holds at most one eclipse of each kind, and multi-sector baselines of years can be phased only when the period is already known to minutes or the single events are deep enough to be found one by one. Catalogued eclipsing binaries are correspondingly scarce in that period range, and the Gaia orbits that would gain most from an eclipse, which fixes the inclination and with it the companion mass, are the ones TESS searches least well.

Gaia DR3 contains about 800,000 non-single-star solutions for astrometric, spectroscopic and eclipsing binaries (Gaia Collaboration, Arenou et al. 2023). The spectroscopic processing, restricted to single-lined F–K stars brighter than magnitude 12, delivered 181,327 SB1 orbits after filtering (Gosset et al. 2025), and the astrometric processing about 165,500 orbits (Halbwachs et al. 2023a). The SB1 orbits are mostly reliable: of 827 binaries shared with the SB9 catalogue, 655 agree to within 10% in period and eccentricity (Merle et al. 2026), and a comparison with LAMOST and GALAH velocities removed the physically unlikely short-period, high-eccentricity solutions (Bashi et al. 2022). Double-lined solutions fare worse, matching the photometry in period and eccentricity for about half of SB2 eclipsing binaries (Rowan et al. 2023) and 20% of SB2 heartbeat stars, against 85% of SB1 ones (Callahan et al. 2025).

TESS eclipsing binaries are catalogued by Prša et al. (2022; 4,584 systems from sectors 1–26), IJspeert et al. (2021, 2024) and Kostov et al. (2025; 10,001 vetted systems from the full-frame images and about 900,000 unvetted machine-learning detections), and Gaia DR3 adds 2,184,477 candidates from its own photometry (Mowlavi et al. 2023). All favour short periods: the 7,936 new systems of Kostov et al. (2025) have a median period of 3.5 d and a 95th percentile of about 11.6 d. Prša et al. (2022) attribute this to the sector length, which suppresses detections from periods as short as 13 d, and to the box least-squares search, which cannot find single eclipses. Planet searches meet the same limit with single-event methods, estimating the period from one transit's shape (Osborn et al. 2016) or catching a second event

from the ground, which also finds eclipsing binaries such as a 61.8-d M dwarf eclipsing a Sun-like star (Gill et al. 2020); "duotransits", one transit in each of two TESS years, typically have periods above 20 d (Hawthorn et al. 2024). In each case the event is found first and the period afterwards.

Heartbeat stars are eccentric binaries that brighten near periastron through tidal distortion, irradiation and Doppler beaming (Kumar et al. 1995; Welsh et al. 2011; Thompson et al. 2012); some show tidally excited oscillations (Fuller and Lai 2012; Fuller 2017). Kepler supplied the first samples (Kirk et al. 2016; Shporer et al. 2016), and the TESS searches have been at short periods: 23 systems at P = 2.7–20 d (Li et al. 2024a), 180 at $P \lesssim 10$ d (Solanki et al. 2025) and 112 from Gaia spectroscopic orbits, with the SB1 orbits cut at P < 13 d (Callahan et al. 2025). A third body shows in eclipse times through the light-travel-time effect (Irwin 1952); eclipse timing found third bodies in 222 of about 2,600 Kepler binaries (Rappaport et al. 2013; Borkovits et al. 2016), and 403 known eclipsing binaries have a Gaia DR3 solution at a period much longer than the eclipse period (Czavalinga et al. 2023a). Over longer baselines, the Harvard plates, about 500,000 taken from 1885 to 1992 and digitised by DASCH with photometry good to about 0.1–0.15 mag (Laycock et al. 2010; Grindlay et al. 2012; Tang et al. 2013; Williams 2025), have been combined with TESS eclipse times to follow a binary's period over a century (Zhang et al. 2026).

New variables are registered in the AAVSO International Variable Star Index (VSX; Watson et al. 2006); times of minimum from AAVSO observers appear regularly in this journal (e.g., Samolyk 2022), and eclipsing binaries found in survey and archival data have been reported here, entered in VSX and characterised with the authors' own photometry (Romanov 2021; De Elías 2025). Guinan et al. (2012) set out the follow-up that eclipsing binaries from space missions need from observers.

Orbits and eclipses have been combined in both directions. The EBLM programme measured orbits for 118 binaries first seen as shallow WASP eclipses (Triaud et al. 2017), and joint fits of this kind supply the masses and radii in DEBCat (Southworth 2015). Starting from Gaia, Rowan et al. (2023) measured masses and radii of 122 stars from SB2 orbits and ASAS-SN light curves, Schmidt et al. (2023) matched SB1 orbits to TESS Objects of Interest, and Halbwachs et al. (2023b) selected 41 bright astrometric binaries that could eclipse. The conjunction prediction itself is not new: Tarrants and Mendes (2023) applied it to Gaia DR3 SB1 elements to vet known TESS and K2 planet candidates and called it their least reliable test, and Yamaguchi et al. (2024) applied it to one 358-d orbit in search of self-lensing pulses. Eclipses in Gaia orbits have been found by other routes: Callahan et al. (2025) found 18 among their 112 heartbeat candidates with a periodogram, El-Badry et al. (2026) found 25 among the orbits of compact-object candidates, 14 of them hierarchical triples with an eclipse period much shorter than Gaia's, and Beck et al. (2024) searched the six oscillating red giants whose Gaia astrometric inclinations are within 3° of edge-on, a sample that includes our A60, which they did not flag. The search itself, examining TESS only inside the windows that Gaia DR3 spectroscopic orbits predict, was taken two weeks before this work began. Chopard-Vilhem (2026a) deposited on Zenodo on 2026 September 27 a survey of 9,389 Gaia DR3 SB1 orbits whose conjunctions are predictable to within a day, in up to three TESS sectors each, reporting 93 eclipsing systems at periods of 13–119 d, 90 of them not previously known to eclipse, with masses and radii for 54. Its object list is embargoed until the accompanying paper is accepted, so the overlap with the systems reported here is not yet known; his period range contains ours, and some overlap is likely. A second record (Chopard-Vilhem 2026b) applies the same test to 1,726 Gaia astrometric orbits within 1° of edge-on. Its latest open version (1.2.0) retains seven surviving detections; its ten dossier entries include withdrawn or known systems and are not ten new discoveries. A comparison against all 235 rows of our object ledger, using positions within 20″ in the parent sample and exact TIC identifiers in the dossier, shares one claimed detection (HD 170829, our A20, found by both searches with the same period). A73, A74 and the withdrawn TYC 3707-251-1 are present in the parent sample; their previously noted pilot-window or event correspondences are not shared claimed detections. All 74 physically confirmed R-labelled and returned additions have exact Gaia DR3 identifiers available; none matches the open parent list or dossier, and none matches a dossier TIC identifier. The separate 93-system spectroscopic list remains embargoed, so its overlap is unknown. Neither record was found by the earlier web and arXiv searches, which did not include Zenodo. The dated comparison and its matching limits are retained in `reviews/novelty-reconciliation/README.md`. This paper claims no priority for the method; its contributions include the objects, archival recoveries and contradicted Gaia orbits.

The work reported here began from a planet question. A public report of two transiting-planet candidates around the K dwarf TIC 4206066 (Rabtsevich 2026) led to a transit search of about 3,100 nearby M dwarfs in TESS full-frame-image light curves (Section 2.3). That search found no new planet (Section 4.6), recovered both reported candidates without using the reported periods or epochs, and turned up four uncatalogued eclipsing systems (A1–A4), two of which had Gaia DR3 evidence of a stellar companion that had never been combined with the TESS photometry. We therefore inverted the search: starting from Gaia's spectroscopic orbits, we tested each for eclipses in TESS. For orbits shorter than 20 d this is a period search at Gaia's period (Section 2.4), and it found 42 uncatalogued eclipsing systems.

For longer periods a Gaia orbit still fixes the conjunction times: the period, the time of periastron, the eccentricity and the argument of periastron together determine when each conjunction falls, to about 2 d at TESS epochs for the median orbit in our sample. Eclipses can only happen at conjunction, so the search can be confined to the predicted windows and use a single-event statistic instead of a period search over years of data. The number of trials falls from a four-year period grid to the windows, which hold about half of the usable data for the median star, and an eclipse that lands on the predicted

conjunction is attributed to the star with the Gaia orbit, which is the hardest step in any blended-field TESS search. We call this eclipses by appointment (Section 2.5). It accounts for 107 of the 470 systems in this paper.

The contributions of this work are (i) the measured appointments-search yield, about 1.7 new eclipsing binaries per hundred Gaia orbits searched at G < 11; (ii) 470 systems with physical eclipse evidence, with ephemerides where constrained (Table A1 and Appendix C); (iii) eleven catalogue corrections and twenty Gaia DR3 orbit solutions that the eclipses contradict; and (iv) a null planet result with its expected yield estimated under stated assumptions, and the reason it was null. "Uncatalogued" means absent from the seven catalogue groups specified in Section 2.7; literature searches are a separate, limited check, not unseen by any pipeline: 72 of the first 104 carry an automated threshold-crossing event of the SPOC or TESS-SPOC pipelines (Jenkins et al. 2016; Caldwell et al. 2020) that was never vetted or promoted to a TESS Object of Interest, and a full-text search of NASA ADS found a prior mention of eclipses for only one (A7; Bryant et al. 2023). Physical classification and catalogue novelty are separate. The sample contains 439 confirmed and 31 probable systems; fourteen plausible signals in the review registers are excluded. Stable keys R001-R216 C01-C72 and P001-P026 preserve the individual review keys and current verdicts (Appendix C). The object labels (A1–A204, B1–B11, C1–C3) were assigned in this work and are used as identifiers throughout; the A-labelled per-object entries with their light curves are in the Supplement, while the individual physical reviews are in the repository, and the data and code are in the deposit described in Appendix B.

2. Data and methods

2.1 TESS photometry

All photometry is from TESS (Ricker et al. 2015), retrieved from MAST. The primary source was the Quick-Look Pipeline (QLP; Huang et al. 2020; Kunimoto et al. 2021) full-frame-image light curves, at 30-min (sectors 1–26), 10-min (27–55) and 200-s (56 onward) cadence. For the long-period work we used the raw aperture flux SAP_FLUX (Section 2.8). Independent reductions of the same pixels served as cross-checks: SPOC 2-min (Jenkins et al. 2016), TESS-SPOC (Caldwell et al. 2020), TGLC (Han and Brandt 2023), eleanor (Feinstein et al. 2019), GSFC-ELEANOR-LITE (Powell et al. 2022), TASOC (Handberg et al. 2021), CDIPS (Bouma et al. 2019), TARS (Boyle et al. 2026), T16 (Hartman et al. 2025) and TEQUILA (Ogunwale et al. 2025). Times are BTJD = BJD_{TDB} − 2457000.

2.2 Gaia DR3 orbits

Orbits are from the Gaia DR3 (Gaia Collaboration, Vallenari et al. 2023) table nss_two_body_orbit (Gaia Collaboration, Arenou et al. 2023; Halbwachs et al. 2023a; Gosset et al. 2025): the spectroscopic types SB1, SB2, SB1C, SB2C and AstroSpectroSB1, and the astrometric Orbital type (Section 2.6), matched to the TESS Input Catalog (TIC 8.2; Stassun et al. 2019; Paegert et al. 2021) by Gaia source identifier.

Gaia gives the time of periastron T_p relative to J2016.0 = BTJD 389.0. Conjunction A is at true anomaly ν = 90° − ω and conjunction B at ν = 270° − ω, with ω the argument of periastron of the star Gaia tracks, which is behind its companion at A. From ν follow the eccentric anomaly E, the mean anomaly M = E − e sin E, and

$$t_c = T_p + 389.0 + M\,P / 2\pi \ \ (BTJD),\ \text{plus integer multiples of } P.$$

Uncertainties were propagated by Monte Carlo through Gaia's errors on P, T_p, e and ω; the conjunction tests used Gaia's full covariance matrix. For circular solutions ω = 0 was assumed. “Primary” denotes the deeper eclipse; a transit is the smaller star crossing the larger, an occultation the smaller star hidden.

2.3 Transit searches of nearby M dwarfs

The parent sample was the TIC 8.2 dwarfs with T_{eff} 3200–4300 K, Tmag 8.5–11 and distance < 40 pc, less every star with a TESS Object of Interest (TOI) or community TOI and every star with a TIC contamination ratio of 0.05 or more; 1,299 of these had 3–8 QLP sectors. Further searches covered fainter and smaller stars, crowded fields, single transits, periods of 0.17–0.5 d and 78 known planet hosts. The searches used box least-squares (BLS; Kovács et al. 2002), with completeness measured from injected transits. Surviving peaks faced odd/even tests, a cyclic-roll null that keeps red noise and destroys transits, difference-image centroids in two field orientations, and archive queries.

2.4 Period folds of Gaia orbits with P < 20 d

The first sample was every Gaia DR3 spectroscopic orbit with P < 20 d and parallax > 10 mas: 736 science orbits, plus 14 Gaia eclipsing binaries as controls, all recovered. A coherent BLS over the flattened QLP flux searched ±max($5\sigma_P$, 0.1%) around P_{Gaia} and $P_{Gaia}/2$, scoring each star with a z value against 300 random periods on the same star. Triage removed VSX matches, rotators and ellipsoidal variables, and required an eccentric system's eclipse to land on Gaia's predicted conjunction.

The same code then ran on orbits with parallax ≤ 10 mas or missing. At G < 9, an independent re-check kept seven (A8–A12, B6, B7). At 9 ≤ G < 10, 2,559 science targets among 3,022 orbits were searched, and 199 of 218 controls (91%) reached z ≥ 5, which measures the search's completeness. After triage and

catalogue searches, 38 folds showed eclipse-shaped dips and 37 were re-analysed to the full protocol of Section 2.7. About 175 eclipse-like stars with a VSX entry within 30″ and nine marginal dips were not pursued, so this yield is a lower bound.

The fold was then extended to $10 \le G < 11.5$ in batch (8,492 and 7,470 orbits in the two subsamples; 6,919 and 6,870 searched; 479 of 549 controls, 87%, at $z \ge 5$ in the first). Here the per-object re-analysis of Section 2.7 was replaced by the same tests run in bulk: triage and the catalogue searches as above, a visual read of every fold, a search of the vetted eclipsing-binary tables, the conjunction test on Gaia's orbit (offset within 0.03 in phase), per-sector depth consistency, and a difference image of the eclipse deficit from TESScut relative to the target and every Gaia neighbour. A review with separately written code then queried ExoFOP, VSX and SIMBAD for every survivor and re-folded 60 of them, drawn at random, from freshly downloaded QLP flux with a blind BLS over 0.5–25 d. The initial survivors were listed as candidates until they had had the per-object re-analysis of Section 2.7; the eight that the difference imaging had flagged and the 60 shallowest of the others (to 28 ppt) have had it, and the 66 it confirmed are counted (A119–A179 and A200–A204, Section 3.2). Two limits of the batch tests showed in that pass: the conjunction test cannot tell an eclipse from the tidal minimum of a heartbeat star when the conjunction falls near periastron (two of the 35 were heartbeat stars, Section 3.3; in four more the conjunction fell near periastron and the eclipse model won outright, and in two a real tidal signal and a real eclipse coexist), and the difference-image offsets select crowded fields and pixel boundaries rather than off-target signals (seventeen of seventeen examined were artefacts: of crowding, of a target sitting on a pixel boundary, of an uncalibrated pixel scale with the measuring box centred on the rounded pixel, of a half-period stack that mixed two unequal eclipses, or of out-of-eclipse windows that fell inside a long eclipse).

The spectroscopic-orbit period folds also extend to the fainter Gaia SB1 population at G 12.75–13 and Gaia $P < 20$ d. The selection stratum contains 2,346 orbit rows; the reviewed subset is not a complete census of that stratum. Its 35 included systems have Gaia $G = 12.753$–12.993 and selection-orbit periods 1.867–19.643 d. These Gaia values describe selection, not newly measured eclipse ephemerides or proof that the eclipses belong to the Gaia orbit. Individual raw-flux, timing, alternative-reduction, attribution and catalogue checks use the same criteria as the brighter period-fold populations.

2.5 Eclipses by appointment

Sample and windows. Gaia DR3 holds 11,446 spectroscopic orbits with $20 \le P < 60$ d and $G < 12$. We searched three subsamples, $G < 10$ (1,422 stars, including 40 Gaia eclipsing binaries as controls), G 10–11 (2,327 stars) and, during the analysis, G 11–11.5 (2,608 stars): 6,357 stars in all. For each orbit the conjunction times and their uncertainty σ_T were computed at every TESS cycle (Section 2.2). Each window had a half-width of $\max(3\sigma_T, 0.25\ \mathrm{d})$, capped at 7 d.

Statistic and cuts. In each window the QLP SAP_FLUX (15-min bins) was searched with a single-event box of 1–14 h against the mean of two side windows 0.75–2.5 durations out. Two shape tests followed: the far ratio (the depth against a baseline 3–6 durations out, divided by the near depth, about 1 for an eclipse) and a deficit in both halves of the box. The null was the same statistic in 40 random windows on the same star outside every predicted window, giving $z = (\mathrm{SNR} - \mathrm{median})/\mathrm{MAD}$, a rank against the star's own null rather than a Gaussian significance. A candidate needed $\mathrm{SNR} \ge 8$, a far ratio in [0.6, 1.6], the event within $3\sigma_T$ of a conjunction, no null window reaching it, and $z \ge 8$ where a null exists. Of the 40 controls, 32 showed an eclipse at the predicted conjunction.

Coverage. Over the 3,708 science stars of the first two subsamples the windows hold a median 51% of the usable cadences, and every cadence for 165 stars (4.5%), where no null exists; for 591 more (16%), z is undefined because all null windows land on the same feature (MAD = 0). Seven of the 54 reported systems therefore never faced the z test (A37, A40 and A42 with no null; A27, A51, A54 and A64 with MAD = 0) and passed on the remaining cuts. One, A18, failed a cut (far ratio 2.19); it was taken from the raw SNR ranking before the cuts were written and rests on six eclipses and its re-analysis.

Classification. Candidates passing the cuts were matched against VSX, ASAS-SN, Prša et al. (2022), IJspeert et al. (2024), Kostov et al. (2025) and SIMBAD; a VSX entry of a non-eclipsing type did not remove a star. The remaining 161 ($G < 10$) and 280 (G 10–11) stars were classified visually. Every star classified as an eclipser, and every unusual non-eclipse signal, was re-analysed independently (Section 2.7), with three exceptions (Section 3.1): 16 at $G < 10$ and 54 at G 10–11. At G 11–11.5, 408 stars passed the cuts, 29 of them catalogued eclipsing binaries recovered blind, and the 379 others were classified visually: 47 eclipsers and three possible ones. Forty-five of these 50 candidates have individual reviews: the original twenty counted systems (A180–A199), followed by 25 historical reviews. The latter comprise one known recovery, one retained plausible signal and the 23 systems reconciled here using the bounded checks of Section 2.7. This reconciliation supports 22 confirmed additions and one plausible signal. Five candidates remain unreviewed and are excluded.

2.6 Edge-on Gaia astrometric orbits

Of the 6,652 Gaia DR3 astrometric orbits with $P < 300$ d and $G < 13$, we kept the 191 whose inclination, recovered from the Thiele–Innes coefficients, is within 2° of edge-on; the inclination served only to rank. For the 177 with QLP light curves, raw SAP flux was searched around Gaia's period and its half, and separately with a per-sector box search over 0.3–13 d and an inspection of the deepest single events.

Underexplored selection and observational gaps. A frozen queue of 40 remaining screened period-fold candidates was reviewed in increasing Gaia G, covering G = 11.5033-11.6584 within the G 11.5-12 selection stratum. The same bounded raw-flux, measured timing, source-attribution, alternative-reduction and Gaia-geometry checks support 39 confirmed and one probable system; the probable TIC 253126207 retains unresolved attribution after a completed review. Periods come from measured eclipse midpoints and same-kind cycle counts where depths alternate. Predicted conjunction centers are search guides, not measured timing points. Timing errors retain their individual meanings, including scatter inflation where specified. Operational processing groups are provenance only.

An independent exploratory pilot examined underexplored windows using existing methods. The appointments branch was adaptively extended from 48 to 96 selected science targets: 90 had TIC matches, 89 had usable QLP sectors, six lacked a TIC match and one matched target lacked usable QLP data. It produced two confirmed new systems and one known IJspeert et al. (2024) recovery, retained as a control. The near-TESS-orbital-period branch examined 72 targets (including the initial 24), retaining two probable eclipsing systems because capable close neighbours prevent definitive attribution. Sixteen long-duration targets gave no promoted case; availability queries for eight prior single-event targets found no new sectors. Failed and unavailable attempts remain in the denominators. Adaptive selection and stopping make these pilot counts unsuitable as unbiased occurrence, completeness or yield estimates. The target selections, attempted observations and outcomes are retained separately from the frozen 40-target queue.

2.7 Independent re-analysis and catalogue checks

Original all-event re-analysis. In the original analyses, every candidate from the appointments search, the G 9–10 fold and the edge-on search was re-analysed independently, with separately written code and with the aim of refuting the claim (Section 6). Each original re-analysis re-measured every eclipse from raw SAP flux, including flagged cadences; fitted a linear ephemeris to all events, so that the period comes from the eclipses themselves; located the flux deficit in difference images relative to the target and every Gaia neighbour bright enough to produce it; re-measured the eclipses in at least one independent pipeline; compared the eclipse geometry with Gaia's orbit and mass function; and predicted the next eclipses.

Conjunction-selected evidence reconciliation. The 23 later-reviewed G 11–11.5 systems were assessed from the deciding raw-event, timing and coverage, pixel attribution, alternative-reduction and Gaia geometry evidence. Existing validated evidence was reused with hashes; missing material checks were completed where practical, including a repeated eclipse in newer TESS coverage for TIC 12864953. Alternative reductions of the same photons test extraction and processing, not independent observing occasions. This bounded reconciliation is not a new exhaustive all-sector search. A completed review can retain a plausible physical outcome. Reported timing errors retain their source-specific meaning: the KELT+TESS period error for TIC 277542587 is heuristic, and the conservative allowance for TIC 12864953 is not calibrated coverage. Reference epochs are named by eclipse kind rather than assumed to identify the deeper primary. The October 31 catalogue checks cover the same seven scoped groups; the cached Chopard v1.2.0 exact-identifier comparison found no overlap among these 23, without asserting a newer online release was searched or resolving the 93 embargoed identities.

Catalogues. Each object was cone-searched in VSX (Watson et al. 2006), SIMBAD (Wenger et al. 2000), the Gaia DR3 eclipsing-binary (Mowlavi et al. 2023), classifier (Rimoldini et al. 2023; Eyer et al. 2023) and non-single-star tables, Prša et al. (2022), IJspeert et al. (2024), Kostov et al. (2025) and ASAS-SN (Shappee et al. 2014; Kochanek et al. 2017; Jayasinghe et al. 2018), through VizieR (Ochsenbein et al. 2000), with a known eclipsing binary as positive control. A repeat on 2026 October 13, adding the GCVS and NSV (Samus' et al. 2017), ASAS-3 (Pojmański 2002), ATLAS (Heinze et al. 2018), ZTF (Chen et al. 2020) and Catalina (Drake et al. 2014), found no eclipsing-type entry for any counted A-object. A full-text query of the NASA ADS index for 248 identifiers of the 74 A-objects (2026 October 12) returned one prior report of eclipses (A7; Bryant et al. 2023); a second (October 13) covered 242 identifiers of the 35 G 9–10 systems.

Status words. **Uncatalogued** means that the eclipses were absent, at the time of the original checks (2026 October 11–12) or the extension reconciliations (2026 October 30 and 31), from VSX, the SIMBAD primary classifications, the Gaia DR3 eclipsing-binary solutions, Prša et al. (2022), IJspeert et al. (2024), the vetted tables of Kostov et al. (2025) and ASAS-SN; an entry only in the unvetted machine-learning list of Kostov et al. (2025, table 2) leaves an object uncatalogued. **Confirmed**: the system passed every check above that its data allow (repeated events in raw flux, a period from the eclipses, a difference image on the target, a second pipeline, a geometry consistent with Gaia's orbit). **Probable**: the eclipses are real and repeat, but one check is missing or failed. **Plausible**: a single event, or two with the period untested. **By-catch**: a real variable on the target whose signal does not follow the Gaia orbit that selected it.

2.8 Known systematics

QLP's detrended flux suppresses long eclipses in post-sector-56 files: it removed the 20-h primary of A16. QLP's sector-80 light curves run 2.7 min late against SPOC and TESS-SPOC in the two systems where this was measured (A11, A98); whether the offset applies to the whole sector has not been tested. At 30-min cadence QLP mid-times can be off by minutes: in A92, sectors 14 and 15 are about 6 min late against three independent pipelines. Depths are those measured in the named pipeline and are diluted

by any neighbour it does not correct for. Because the QLP aperture changes between sectors, a neighbour can make one sector's eclipse 9–27% shallower than the others (for example A85), so depths are taken from the sectors that agree with difference images or a second pipeline.

Individual review uses the same five deciding checks on representative available events, rather than requiring all-event or all-sector extraction. Review of the G 10–11.5 period-fold candidates includes raw-flux events and uncertainties, timing and coverage, each source's own Gaia covariance, alternative reductions, neighbours and pixel attribution, and physical plausibility. Tested analysis primitives and valid existing evidence are reused where appropriate; these reviews are not all independent-code replications. Alternative reductions, including custom TESScut aperture extractions, share the same TESS photons and test reduction robustness rather than independent astrophysical detections. Original cycle counts and timing baselines are retained in each report; no uniform precision period errors are inferred from point estimates. Physical verdict, catalogue novelty, review limitations and next tests are recorded separately. All included systems meet the seven-group scoped catalogue standard; known recoveries are excluded from unique additions. The unified scientific review register links physical outcomes and their evidence (Appendix C). Successful shared checks are retained, and failed requests are not negative matches. The audits and row-level coverage are in `reviews/novelty-reconciliation/README.md` and `reviews/2026-10-30-next70/`. Literature searches do not establish exhaustive priority.

3. Results

3.1 Sample and yield of the appointments search

Table 1 gives the yield of the appointments search stage by stage. In the G < 10 subsample, 181 stars passed the cuts. In the G 10–11 subsample, 447 of 2,327 stars passed, and the 280 that no eclipsing-binary catalogue lists were inspected. Of the 70 stars re-analysed independently, 54 are eclipsing systems: 50 confirmed and four probable (A37, A40, A53, A68). Three are new heartbeat stars (Section 3.3). Table 2 lists the rest. Four candidates were refuted, three remain plausible, three on-target stars were already known, and 20 catalogued eclipsing binaries were recovered blind in each subsample. In the G 11–11.5 subsample, 408 of 2,608 stars passed the cuts; 29 were catalogued eclipsing binaries recovered blind, and of the 379 inspected, 47 were classified as eclipsers and three as possible. The original twenty strongest are confirmed eclipsing systems (A180–A199). Of 25 later historical reviews, one known recovery and one plausible signal remain outside the counted sample; the other 23 have now been reconciled under Section 2.7, supporting 22 confirmed additions and one plausible signal (TIC 8853239). Thus this stratum contributes 42 counted confirmed systems. Forty-five of the 50 candidates have individual reviews, and five remain unreviewed.

The automated search missed whole eclipses in two ways. The search first discarded 0.5 d after every data gap, which removed whole eclipses (five of the six of A18); the trim was reduced to 0.1 d and the G < 10 subsample was searched again. And in many systems whole eclipses lie in cadences that QLP flags, mostly for scattered light, although the flux is usable: five of the ten eclipses of A30, for example. These were recovered from flagged cadences, other pipelines or cutout photometry.

3.2 The eclipsing systems

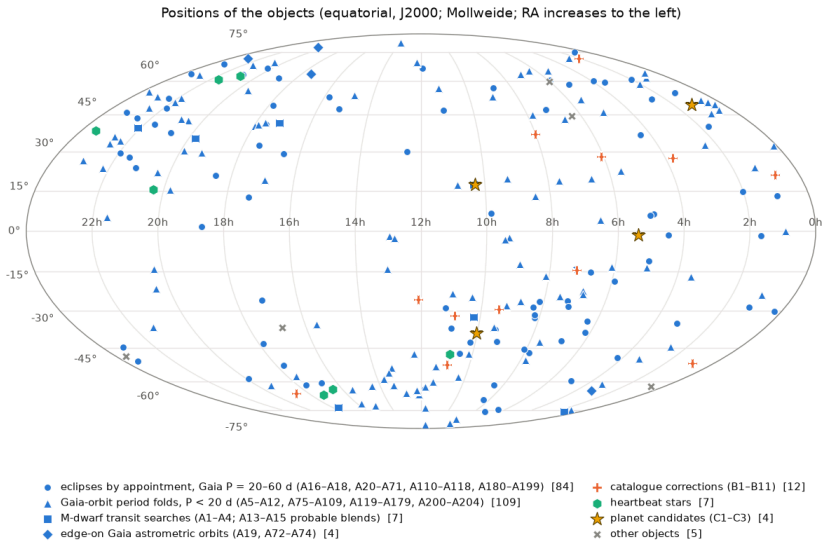


Figure 1. Positions of the A-labelled catalogue-screened systems and comparison objects (equatorial, J2000, Mollweide projection). Marker shape gives the search that found each eclipsing system; the catalogue corrections, heartbeat stars, planet candidates and other objects are shown separately. Counts are in brackets; the R-labelled and returned C-labelled systems of Appendix C are not plotted.

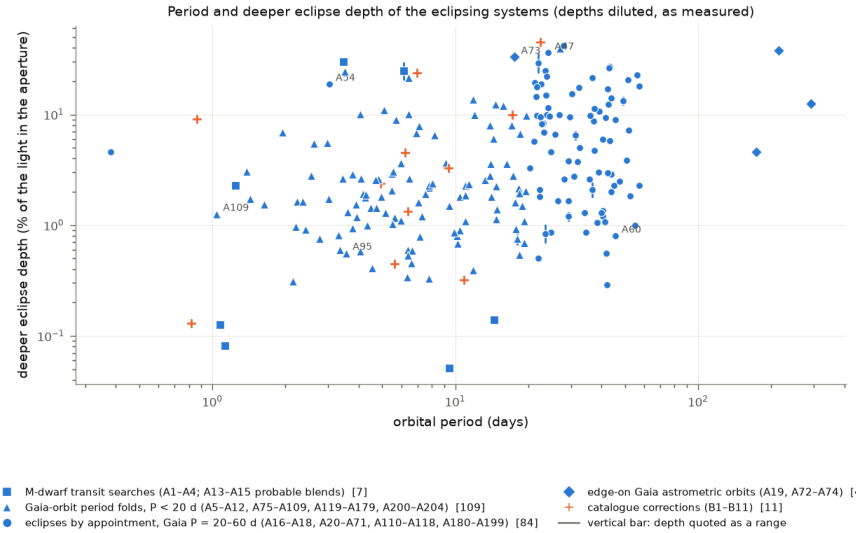


Figure 2. Orbital period and deeper eclipse depth of the A-labelled systems and the catalogue corrections; R-labelled and returned C-labelled systems are not plotted. Depths are as measured in the pipeline aperture, diluted by any blended light; a vertical bar spans a depth quoted as a range. The appointments search (circles) fills the 20–60 d range that the period folds (triangles) do not reach.

The physical sample contains 470 systems (439 confirmed and 31 probable), all meeting the stated seven-group uncatalogued standard. Fourteen plausible signals in the review registers are excluded. By search method, 107 systems come from predicted spectroscopic conjunctions, 355 from Gaia-orbit period folds, four from the nearby-M-dwarf-selected search (including neighbouring hosts), and four from near-edge-on Gaia astrometric orbits. Physical classification, association with a Gaia orbit and catalogue status are separate assessments.

HIP 1060 (A47) illustrates the method (Figure 3). It shows a 34%, 6.9-h total eclipse at conjunction A of its Gaia double-lined orbit and a 36%, 16-h annular transit at B, eleven events over eight years at P = 24.10512 d. Because Gaia measured both velocities, one model gives both masses (1.10 and 1.22 $M_\odot$) and radii (1.26 and 1.83 $R_\odot$). Three eclipses in sector 106, predicted before those data were examined, fell within 1.5 min of the ephemeris. HD 170829 (A20), at G 6.3 the brightest star here, is a G8 subgiant with an asteroseismic mass, eclipsed every 26.384493 d; VSX lists it as NSV 24480 with type “VAR:” and no period.

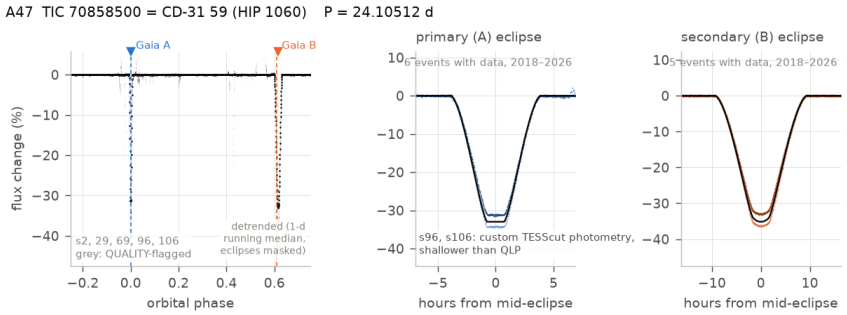


Figure 3. HIP 1060 (A47, TIC 70858500). Left: QLP SAP flux, detrended with a 1-d running median with the eclipses masked, folded on P = 24.10512 d; dashed lines mark Gaia's predicted conjunctions A and B. Middle: the six 34%, 6.9-h total eclipses at A, 2018–2026. Right: the five 36%, 16-h annular transits at B.

Appointments. Of the 54 systems, 52 are long-period eclipsing binaries found at Gaia's conjunctions, with periods from 20.29 d (A35) to 57.24 d (A63). Two are by-catch at periods other than Gaia's: A45 (31.15 d, against a Gaia double-lined period of 21.68 d) and A54 (a 3.02-d eclipsing pair whose 39-d Gaia orbit is spurious). A third by-catch object, A46, is a 0.38-d contact or ellipsoidal variable on an unresolved pair and is not counted. A180–A199, from the G 11–11.5 subsample, have periods from 21.68 d (A181) to 56.17 d (A198). Ten have giant or subgiant primaries, and in five of these the deeper eclipse is the occultation of a smaller, hotter companion (A182, A183, A189, A192, A199); three carry VSX rotational classifications that describe the primary's spot modulation and miss the eclipses, so EA is added (A184, A185, A194), and a fourth rotational entry is replaced (A192); A191 is WASP0-TR-06, a single partial event of 2000 rejected as a planet (Kane et al. 2005), which now has its period; two community planet candidates on ExoFOP are the occultations of stellar companions (A197, A199); the Kostov et al. (2025) flag near A196 sits on a neighbour 37″ away; two show one eclipse kind only, as their geometry requires (A194, A199); and the orbital tensions stop short of the rule, the largest being A188's occultation-to-transit spacing, 2.9σ from Gaia's, where the occultation times also wander by about 50 min against the transits in a way that neither apsidal motion nor light-travel time explains, and A196's $e \cos \omega$ at 2.3σ, where APOGEE velocities side with the eclipses.

The later conjunction-selected systems retain individual limitations: TIC 277542587 combines repeated KELT eclipses with TESS at a heuristic working period precision of 0.0002 d; TIC 12864953 has a repeated secondary in sector 107 and a conservative 0.00015-d timing allowance, while its new primary window is quality-flagged and a phase-separation tension of about 3.35σ remains unresolved. Neither has been added to the formal Gaia-orbit contradiction count without the stated comparison-rule analysis. TIC 8853239 retains only one primary and an untested period and remains plausible.

Period folds at P < 20 d. A5–A12 come from the folds at parallax > 10 mas and at G < 9; all eight were known spectroscopic binaries, and the eclipses are new. A5, A6, A7 and A10 are probable, because the fold makes period agreement certain by construction and none has a pixel test or a second pipeline. The fold at G 9–10 produced A75–A109, all 35 confirmed. A93 was already typed as an eclipsing binary in SIMBAD from a SuperWASP flag without a period (Schanche et al. 2019), so 34 are counted. In A95 the eclipse times vary with an amplitude of 3.4 ± 0.3 min on the 541-d period of Gaia's astrometric orbit for the star, consistent with the light-travel time that orbit predicts (chance coincidence ~10%).

Period-fold populations and physical discrimination

The G 10–11.5 period-fold search examined 13,789 Gaia spectroscopic orbits and selected 266 signals. Individual analysis of 242 eclipse candidates supports 230 confirmed targets (including three known catalogue recoveries), nine probable targets, two non-eclipsing heartbeat stars and one plausible signal. The screening register retains known planet-candidate signals, catalogue corrections and historical holds separately. The catalogue-screened additions use the same seven-group standard as the conjunction search.

The fainter Gaia SB1 period-fold population contributes 35 additional systems: 28 confirmed and seven probable. These are detections by the existing Gaia-orbit folding method at periods below 20 d, rather than a separate search technique. Known eclipsing-binary recoveries and a blind recovery of A4 are excluded from additions. Appendix C groups all individual reviews by scientific outcome, with the original identifiers retained solely as evidence keys.

Physical discrimination matters most where a tidal minimum resembles an eclipse or a nearby source contaminates an aperture. Two G 10–11.5 signals, TIC 391918838 and TIC 98149548, are heartbeat stars without eclipses (Section 3.3); A144 combines tidal variation with eclipses. Pixel-level tests resolve apparent centroid offsets caused by crowding, while independent reductions test shallow events and systematic features. Unresolved attribution or incomplete decisive checks retain a probable or plausible classification rather than being absorbed into the confirmed sample. The archived reports preserve fitted depths, event counts, ephemerides, Gaia comparisons and each deciding limitation.

M-dwarf searches. A1–A4 were detected as planet-like signals on nearby M or K dwarfs. A1, A2 and A4 are eclipses on a neighbouring star whose light enters the dwarf's aperture. A3 is on the target and is probable: Gaia shows that its companion is most likely a star, but the companion's orbital period is not known.

Edge-on astrometric orbits. A19 is probable: three eclipses repeat at a period that agrees with Gaia's to 0.7σ, but they fall about 2.5σ from the nearer predicted conjunction, and two exist only in flagged or QLP-only data. A72 eclipses on its 215.0116-d Gaia orbit itself. A74 is probable, with two 4.6% TESS eclipses three cycles apart and a ZTF night at the predicted time of the cycle between them. A73 (TYC 4590-364-1) is a 17.54-d eclipsing binary inside a 125-d Gaia orbit. Its eclipses have faded from 33% to 13% in four years and their timings vary by hours (Figure 4). NEOWISE W1 photometry, read during the analysis, has the primary no deeper than 20% in 2016–17 and at least 21% in 2021 August, so the depth rose before it fell. A third body is needed to explain this, though the data do not show that it is Gaia's.

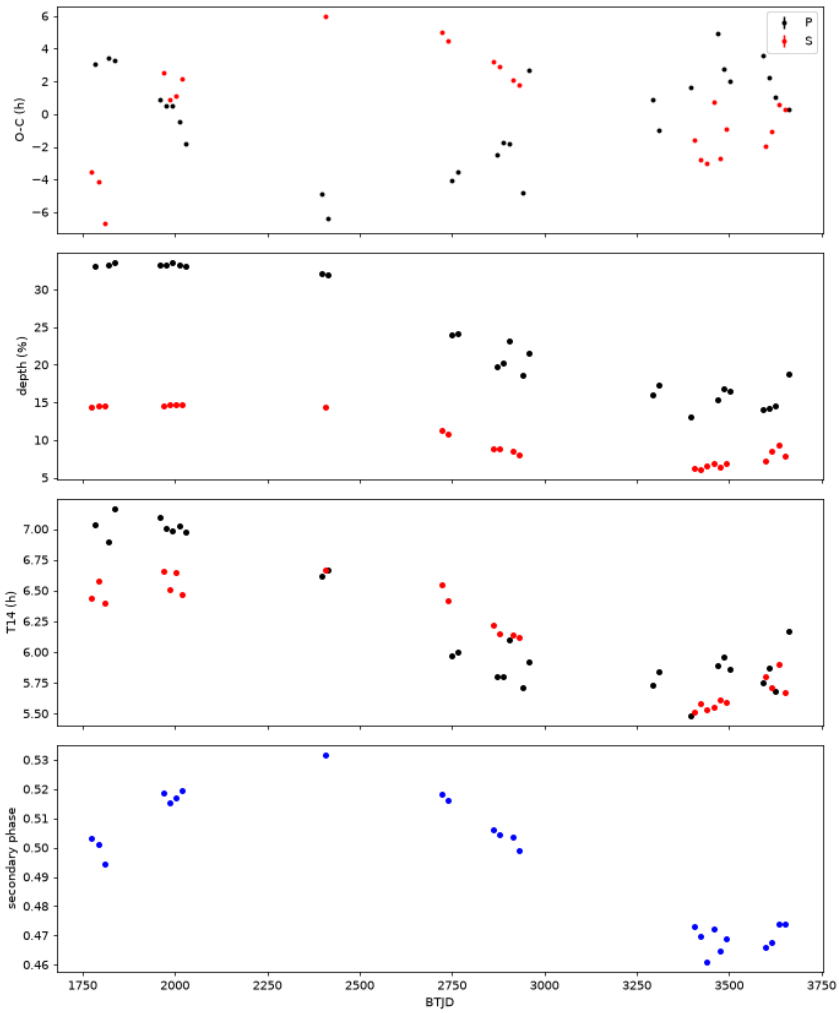


Figure 4. TYC 4590-364-1 (A73), from fits to individual primary (black) and secondary (red) eclipses in QLP sectors of 2019–2024. From top: timing residuals against a linear 17.54-d ephemeris, eclipse depths, total durations, and the phase of the secondary (blue). The eclipses fade from 33% to 13% and narrow, and the timings vary by hours.

The G 11.5-12 spectroscopic-orbit period-fold population contributes 40 reviewed systems (39 confirmed, one probable), and the near-orbital-period pilot contributes two probable systems. The probable cases retain unresolved target-versus-capable-neighbour attribution. TIC 397586559 retains Gaia drift limitations; TIC 343926224 excludes anomalous windows from timing and uses a scatter-inflated uncertainty; TIC 314041837 has localized eclipse cores atop modulation, and its fit window is not a physical eclipse duration. These limitations do not convert predicted conjunctions into timing measurements or independent reductions into independent observations. The fresh conjunction pilot contributes TIC 382042754 and TIC 389128842 as confirmed systems; TIC 155303130 is a strong known recovery and adds nothing. These findings demonstrate useful underexplored selection windows, not previously untouched parameter space or an unbiased pilot yield.

Physical classification and limitations. The physical sample contains 470 systems (439 confirmed and 31 probable), all meeting the stated seven-group uncatalogued standard. Fourteen plausible signals in the review registers are excluded. By search method, 107 systems come from predicted spectroscopic conjunctions, 355 from Gaia-orbit period folds, four from the nearby-M-dwarf-selected search (including neighbouring hosts), and four from near-edge-on Gaia astrometric orbits. Physical classification, association with a Gaia orbit and catalogue status are separate assessments. Repeated discrete eclipses, independent reductions and source attribution distinguish secure eclipsing systems from fold-only or ambiguous signals. A confirmed on-target eclipse does not by itself establish its relationship to a discrepant Gaia orbit. Report-specific attribution, secondary-localization and reduction limitations remain explicit in Appendix C. The same criteria apply throughout this continuous version 1 analysis; the current reconciliation retains new deciding measurements and changed classifications in the dated evidence.

Probable blends. Three signals from the crowded-field and known-host searches, A13–A15, are eclipses on faint stars near the targets, which the pixels locate but not to a single star or not with enough sectors. They are listed as probable and are not counted among the 199.

The newly reviewed TIC 311918550 retains robust target-eclipse attribution and conjunction agreement despite a reported 8.56σ period difference from Gaia. This is a newly flagged orbital tension; it has not been added to the twenty contradictions without the same formal comparison-rule analysis.

3.3 Heartbeat stars

Nine eccentric binaries show a brightening and dimming locked to periastron rather than an eclipse (Table 3), and a tenth shows both. For each we fitted the equilibrium-tide model of Kumar et al. (1995) to the light curve alone and compared the e and ω it returns with Gaia's radial-velocity orbit. Three come from the appointments search. HD 193389 (Gaia P 20.7966 d) recovers Gaia's ω to 0.3° and e to 0.016 (3.2σ by Gaia's error); TYC 8203-301-1 also shows a coherent 842-ppm oscillation at 23.0009 ± 0.0006 times the orbital frequency, which we read as a tidally excited oscillation. Four more come from the G 9–10 fold. Of nine light curves set aside there for a brightening-and-dip feature, seven put it at Gaia's periastron (about 3 in 10,000 by chance); three of the seven are published heartbeat stars, recovered blind, and four are in no catalogue or heartbeat search. BD+61 2067, BD+37 4854 and CD-66 2078 are confirmed; TYC 9045-989-1 is a candidate, its ω 20° (2.4σ) from Gaia's. Two more come from the batch candidates of Section 3.2, where the conjunction test had read their tidal minimum as an eclipse. In TYC 3347-1439-1 (TIC 391918838, *P* 8.92 d, Gaia *e* 0.44) the 2.2 ppt dip is about 20 h wide where the geometry allows 6 h of eclipse, and the Kumar et al. model with Gaia's *e* and ω held fixed beats any trapezoid by $\Delta\chi^2 = 766$; in HD 404939 (TIC 98149548, *P* 2.4026 d, a near-twin A-star SB2) the tidal curve reproduces the whole 9.2 ppt dip with no eclipse, the limit at the dip being 0.4 ppt (2σ) with injected eclipses recovered down to 0.5 ppt. Both are uncatalogued. The same blind spot has a constructive side. In TYC 3624-860-1 (A144, *P* 7.3057 d) an 18 ppt peak-to-peak tidal curve with periastron at phase 0.25 carries a 12 ppt M-dwarf transit at conjunction A and a 2.1 ppt occultation at phase 0.3625, 0.26σ from the separation Gaia's *e* and ω predict; the tide alone loses to tide plus eclipses by $\Delta\chi^2 > 11000$, and the batch's eye-read of "messy oscillation and dip" was this. In CD−35 10394 (A150) a tidal dip of about 1 ppt sits under a 15 ppt eclipse 18° past periastron. In four more of the 35 the conjunction fell near periastron and the heartbeat model lost to the eclipse by $\Delta\chi^2$ of 5000 or more (A138, A141, A143, A147).

3.4 Gaia DR3 orbit solutions contradicted by the eclipses

Eclipses fix the period, and their spacing fixes e cos ω, independently of any velocity measurement. We count a Gaia solution as contradicted when its period is spurious or wrong, or when the period or e cos ω differs from the eclipse value by 3σ or more of Gaia's quoted error, whether Gaia's covariance is propagated in full or its correlations are ignored, and however the ephemeris is carried. Conjunction offsets alone, and e and ω from eclipse durations, are recorded as tensions. Nineteen of the 184 counted systems compared with a non-circular Gaia orbit meet the rule, and so does A113, whose circular Gaia solution carries the wrong period (Table 4; twenty in all): three periods are spurious or wrong (A54, A9, A113, the last the 63-d scan-cadence alias of the eclipse period), ten are 3.4–8.0σ off (A110 by 21 min per orbit, 4.5σ; A117 by 87 min per orbit, 3.5σ; A147 by 4 min per orbit, 4.3σ, which the batch's conjunction offset had extrapolated to +0.019 in phase; A167 by 1.05 min per orbit, 8.0σ, an error understated by a perturbed double-lined solution whose two conjunctions still match the eclipses at Gaia's own epoch; A202 by 11 s per orbit, 3.4σ, whose $e\cos\omega$ is 3.5σ off as well, the occultation at phase 0.49993 ± 0.00045 making the orbit circular where Gaia found $e = 0.054 \pm 0.015$), and seven have e cos ω 3.7σ or more off (for A145 the secondary eclipse at phase 0.49996 ± 0.00018 gives 0.0000 ± 0.0003 against the +0.036 ± 0.008 of Gaia's $e = 0.039$, 4.5σ, so the orbit is circular and the eccentricity spurious; for A159 the secondary at phase 0.62842 ± 0.00004 gives 0.2005 ± 0.0002 against Gaia's 0.252 ± 0.013, 3.9σ with Gaia's covariance and 3.3σ without, and Gaia's period is 3.2σ long as well). Three conjunction offsets (A89, A104, A87) move between about 2σ and about 8σ with the treatment and are not counted; A109's period difference falls to 1.8–2.7σ once its observed period change is included.

3.5 Catalogue corrections

Eleven catalogued stars have the wrong type, the wrong period or none (B1–B11, Table 5). Nine corrections are established by the data; two are proposed, B1 because its host is not yet separated from an unresolved neighbour 4″ away and B7 because a contamination ratio of 1.77 leaves a blend possible. V1274 Tau (B5) is typed RS at 6.62 d in the GCVS and 6.448 d in VSX, and ASAS-3 lists it as ESD/EC at 1.727 d (Pojmański 2002). TESS shows 4.5- and 2-ppt eclipses at Gaia's 5.62323 d, and the ASAS-3 photometry itself peaks at the 6.450-d rotation period with no power at 1.727 d. Five appointment systems (A20, A25, A31, A32, A52) are VSX type revisions.

3.6 Planet candidates and other objects

The M-dwarf search recovered both transit candidates of TIC 4206066 (Rabtsevich 2026) blind, at SNR 15 and 10 (C1; P = 11.132690 and 3.182785 d). TOI-5394.01 (HD 89277, C2) is the occultation of the M-dwarf companion in its Gaia SB1 orbit (P 15.195017 d), not a planet transit. The search around known hosts recovered the two TOI-6284 signals and a TOI-1467 threshold-crossing event, all in a preprint (C3; Tschudi 2026). Table 6 lists the other objects, for example GJ 4348, a young M dwarf that is plausibly a near-twin M+M pair at 0.223864 d, a case one spectrum at quadrature would settle.

Table 1. Yield of the appointments search.

	G < 10 subsample (1,422 stars)	*G* 10–11 subsample (2,327 stars)	*G* 11–11.5 subsample (2,608 stars)
Sent to independent re-analysis	16 stars (two more, HD 201073 and HD 7423, were dismissed on inspection before any check)	54 stars (TYC 5976-120-1, already in VSX as an eclipser, was not sent)	45 of 50 candidates have individual reviews: 20 original systems plus 25 later historical reviews; 23 of the latter reconciled here, with one known recovery and one retained plausible signal outside this queue; 5 remain unreviewed

	G < 10 subsample (1,422 stars)	*G* 10–11 subsample (2,327 stars)	*G* 11–11.5 subsample (2,608 stars)
Eclipsing systems, uncatalogued	14 confirmed (A117 after submission, from the corrected-null re-run)	42 confirmed + 5 probable (A110–A116 and A118 after submission, from the corrected-null re-run) (A37, A40, A53, A68) long-period; by-catch: TYC 447-247-1 (A54, 3.02 d) and TYC 7897-48-1 (A45, a 31.15-d eclipsing binary); BD+30 2277 (A46) is a contact or ellipsoidal variable and is not counted	42 confirmed and counted: 20 original (A180–A199) plus 22 after current reconciliation; catalogue status is separate from planet-candidate annotations
Confirmed heartbeat stars, new	1 (HD 193389)	2 (TYC 8203-301-1, with tidally excited oscillations at 22, 23 and 24 times the orbital frequency; BD+59 2405)	none examined yet
Recovered as already known	20 catalogued eclipsing binaries recovered blind (not sent for re-analysis)	20 catalogued eclipsing binaries recovered blind; 3 on target (TOI-1897 and TOI-5149, planet candidates that are M-dwarf eclipsing binaries; TYC 1194-680-1, a SuperWASP EA with the wrong period); 2 known eclipsing binaries on neighbours 63″ away	29 catalogued eclipsing binaries recovered blind; one known recovery in the later 25 historical reviews (TIC 318209162), excluded from additions
Refuted	1 (systematics)	3 (all neighbour contamination at 36–60″; two of the neighbours are uncatalogued eclipsers themselves)	0 of 45
Plausible, unresolved	1 single event (BD+16 2639)	3 (TYC 7687-990-1, whose eclipses appear in 2025 but not 2019–2023; Wolf 73 and HD 282670, single events, the latter not the Gaia companion's)	2 reviewed plausible signals: TIC 47542584 (historical one-sector pair) and TIC 8853239 (one primary, period untested); 5 candidates unreviewed

Table 2. Appointment candidates that are not new confirmed eclipsing binaries.

Object	Outcome	What the check found
Recovered, already known		
TOI-1897 = TIC 420645189	Known: retired planet candidate	TOI-1897.01 and .02 are this system's primary and secondary eclipses, retired by TFOPWG as FP/EB on 2024-09-17; P = 48.533998 ± 0.000022 d. Recovered blind. New: the Gaia SB1 gives f(M) = 0.0088 $M_\odot$, a 0.25 $M_\odot$ M-dwarf companion (K_1 12.4 km/s against about 50 m/s for the putative planet), and a sixth A eclipse in s86
TOI-5149.01 = TIC 286094277 = TYC 3578-1416-1	Known TOI; reinterpreted as an EBLM	A planet candidate since 2022-02-08; the 1.00–1.11%, 9.4–9.8-h transit is the TOI signal, P = 27.37163 ± 0.00006 d. New: a 602 ± 57 ppm secondary eclipse at phase 0.697 (z = 7.1; also in SPOC, TESS-SPOC and TARS); e cos ω = 0.3128 and e sin ω = −0.093 match Gaia; f(m) = 0.0014 gives m_2 = 0.11–0.13 $M_\odot$, R_2 = 0.15 $R_\odot$, T_2 ≈ 3100 K, so the companion is a low-mass star (an EBLM) rather than a planet. Speckle imaging shows a companion at 0.086″ (Δm 1.97)
TYC 1194-680-1 = TIC 435855781 = 1SWASP J004527.74+210427.9	Known eclipser; period and type revision	Listed as EA/EB with P = 1.66 d in Thiemann et al. 2021 (SuperWASP citizen science; not in VSX). TESS gives P = 22.450574 ± 0.000054 d (Gaia SB2 22.44398 ± 0.0033, 2.0σ off); 45% and 39% eclipses; joint SB2 fit: M = 1.144 + 1.161 $M_\odot$, R = 1.93 + 2.13 $R_\odot$, e = 0.1826, ω = 57.4°. This is a revision of the period and type, not a discovery
HD 181068 and seven catalogued EBs	Known; recovered blind	HD 181068, the Kepler triply eclipsing triple, recovered blind, and seven catalogued eclipsing binaries among the first 711 stars triaged: LW Pup, TIC 188625601, four IJspeert et al. 2024 OBAF eclipsers and one Prša et al. 2022 TESS EB. The full rerun of the G < 10 subsample recovered 20 known eclipsing binaries in all
Refuted		
TIC 71350905 = TYC 4361-615-1	Refuted: systematics	The claimed long periastron eclipse was orbit-edge scattered-light ramps: sector 40's first orbit, with no conjunction within 14 d, shows the same 1% drop. The mass function (0.0025 $M_\odot$) makes the companion a ~0.2 $M_\odot$ M dwarf whose transit could be at most 0.14% deep
TIC 212431508 = TYC 5162-1158-1	Refuted: neighbour, itself an uncatalogued eclipser	Both dips (13% and 4.3%) are on Gaia DR3 4230027841458799744 (G 11.02, 35.9″, same parallax, probably a wide companion), which loses 16.1% and 5.3% while the target loses 0.6% and −0.3%; depth falls 13 → 2.2 → 0.7% as the aperture tightens. The neighbour is itself a Gaia SB1 (P 33.294 d) whose A–B spacing is 2.9σ from its own orbit: a separate lead
TIC 17238861 = TYC 2508-895-1	Refuted: neighbour, itself an uncatalogued eclipser	The 1.9% dips are 10.5% and 9.6% V-shaped eclipses of TIC 17238865 = TYC 2508-1098-1 (G 12.20, 60.4″, same parallax but not bound); the target's core pixels change by no more than 0.1–0.4%. The neighbour's period is any of 732.0556/N d (N = 1–28); 48.80 d is one of these 28 aliases, the one tied to the target's orbit
TIC 265677843 = TYC 3335-389-1	Refuted: background eclipsing binary	The 7.37% dip is a 32.4% eclipse of Gaia DR3 250601205600534272 (G 12.06, 37″, parallax 0.85 mas, a background star); the target loses 0.01 ± 0.04%; TARS 0.83% against QLP 7.37%. Attribution to the target would also require a period 3.9σ from Gaia's
Plausible single events		
BD+16 2639 = TIC 334634917 = HIP 70251	Plausible, unconfirmed	One 0.52%, 2.50-h grazing eclipse at BTJD 2676.6639, 0.4σ from Gaia's A conjunction, identical in five pipelines and on the target's pixels, in the only TESS sector (s50). The geometry (i ≈ 86.3°) predicts the 2.5–3.0-h duration and no secondary. An unrelated dip has a chance of a few percent of falling inside a ±0.5 d window. Next A: BTJD 4345.49, 4365.60 (±0.17)
Wolf 73 = HIP 6972 = TIC 365156806	Plausible, unconfirmed	One 2.98 ± 0.10%, 3.72-h flat-bottomed dip at BTJD 2481.9196 in SPOC 2-min data, 0.29σ from Gaia's B conjunction, on target; no other conjunction covered. A grazing occultation of a companion of at least 0.50 $M_\odot$. SPOC's own pipeline flagged the same event in 2021 at a spurious 32.139-d period. Next B: BTJD 4327.91 ± 0.47 (Gaia's period only)
HD 282670 = TYC 2392-1644-1 = TIC 366141684	Plausible, unconfirmed	One 2.10 ± 0.01%, 6.8-h eclipse at BTJD 2491.9694 in QLP and TESS-SPOC alike, on target to 1.1″ (3σ) by centroids and a difference image, with 115 other stars in the sector clean at that time; 0.06σ from Gaia's A conjunction, but **not the Gaia companion**: its mass function allows at most a 0.9% transit, and no companion of the 3.5–5.2 $R_\odot$ F primary fits the duration at any period the other sectors allow. Best read as an eclipse in an unresolved bound F/A third star carrying ≥ 6% of the light (Gaia duplicated_source, RUWE 1.22); period unknown, > 35 d if circular. SuperWASP DR1 holds a second eclipse on 2004-09-13 (−1.8% on two cameras over 3.3 h, a comparison star flat; the TESS trapezoid fitted to it gives depth 2.10% and mid-time JD 2453261.681), so the period is 6230.29/N d; the TESS non-repeat condition, SuperWASP's other nights and 2,000 KELT-North points from 2006–2008 leave 32 integer-N values between 36.9 and 188.8 d, none at Gaia's period. ASAS-SN or Gaia DR4 epoch photometry should cut these to a few, and one ground night at a predicted mid-time settles a candidate; one high-resolution spectrum would settle the third star
Unresolved		

Object	Outcome	What the check found
TYC 7687-990-1 = TIC 271053559	Plausible; eclipses only in 2025	In s89: a 31.33% V-shaped eclipse and two flat 5.19% and 5.51% eclipses, all on the target (difference images, TARS, clean common mode), at Gaia's 24.65-d period (P = 24.6548 ± 0.0018 d). However, the same ephemeris falls on covered, flat data in s8, s9 and s62, and a scan of every period from 24.58 to 24.76 d (the s89 value at ±50σ, Gaia's at ±5σ) finds, for each one, at least one of s8, s9, s35 and s62 covering the A conjunction with unflagged data and nothing deeper than 0.63% there, so the absence before 2025 is not an artefact of the one-sector ephemeris. Gaia also has a 300.8-d astrometric orbit on the star; our hypothesis, which is not proven, is a compact triple whose inner orbit precessed into eclipse between 2023 and 2025; the implied nodal-precession period of a few decades allows it. Next A: BTJD 4327.29, 4351.95 (±0.05; the third body can add hours)

Table 3. Heartbeat stars: nine new (one of them also eclipsing) and one candidate. The equilibrium-tide model of Kumar et al. (1995), fitted to the light curve alone, is compared with Gaia's independent orbit.

Name	TIC	T (mag)	P (d)	e	Origin	Status	Note
HD 193389	51223058	7.063	20.7974 ± 0.0015	0.36	App	new	F5.5 bright giant; the model recovers Gaia's ω to 0.3° and e to 0.016 (3.2σ)
BD+59 2405	405590521	9.886	25.90	0.59	App	new	A2 star in Trumpler 37; P is Gaia's
TYC 8203-301-1	147413927	10.134	22.0440 ± 0.0005	0.68	App	new	tidally excited oscillation: 842 ppm at 23.0009 ± 0.0006 times the orbital frequency, weaker peaks at 22 and 24 times
BD+61 2067	365670788	8.653	8.216284 ± 0.000094		Fold	new	the light curve alone returns Gaia's e and ω
BD+37 4854	333169278	9.38	5.077463 ± 0.000021		Fold	new	Gaia's ω returned once Doppler beaming is added
CD−66 2078	302055860	9.546	5.040480 ± 0.000030		Fold	new	Gaia SB2, near-twin F stars; returns Gaia's e and ω
TYC 9045-989-1	350856983	9.368	4.682234 ± 0.000021	0.12	Fold	candidate	Gaia SB2, near-twin F stars; ω 20° (2.4σ) from Gaia's; largely ellipsoidal
TYC 3347-1439-1	391918838	9.860	8.9222	0.44	Batch	new	the 2.2 ppt batch "eclipse" is a 20-h tidal minimum at periastron, 0.16 d after Gaia's conjunction; Kumar model with Gaia's e, ω fixed, $\Delta\chi^2$ 766 over any trapezoid
HD 339939	98149548	10.107	2.402590 ± 0.000010	0.15	Batch	new	Gaia SB2, near-twin A stars; 15 ppt peak-to-peak tidal curve with its minimum at conjunction A; eclipse limit at the dip 0.4 ppt (2σ), injections recovered to 0.5 ppt; P 3.6σ from Gaia's 2.40241
TYC 3624-860-1	200398115	10.765	7.305664 ± 0.000049	0.21	Batch	new, eclipsing (A144)	18 ppt peak-to-peak tidal curve with periastron at phase 0.25, carrying a 12 ppt M-dwarf transit at conjunction A and its 2.1 ppt occultation at phase 0.3625; the tide alone loses by $\Delta\chi^2 > 11000$

T: TESS magnitude from the TIC. e: Gaia's eccentricity where the text quotes it. HD 193389: Gaia SB1 P 20.7966 d. Origin as in Table A1: the first three were found by the appointments search (Section 2.5), the other four set aside by the G 9–10 Gaia-orbit fold (Section 2.4); Batch, the three found among the shallow candidates of the G 10–11.5 fold by the per-object pass (Section 3.2). None of the first nine eclipses at the fitted inclination; TYC 3624-860-1 does, and is counted as A144 in Table A1.

Table 4. Gaia DR3 orbit solutions contradicted by the eclipses: 19 of the 184 counted systems compared with a non-circular Gaia spectroscopic orbit, and A113, whose circular Gaia solution carries the wrong period. A solution is counted as contradicted when its period is spurious or wrong, or when its period or e cos ω differs from the eclipse value by 3σ or more of Gaia's quoted error under every treatment of its covariance.

Label	Name	Gaia P (d)	Eclipse P (d)	Fails in	Discrepancy
A9	HD 7601	13.42 (SB2)	27.0822013 ± 0.0000180	period (wrong)	$2P_{\rm Gaia}$ differs by 100σ; $(M_1 + M_2)\sin^3 i$ = 0.49 $M_\odot$, impossible for this pair
A12	HD 217492		14.0343984 ± 0.0000110	period	4.2σ (conjunction phase 5.5σ)
A21	TYC 4244-623-1		30.763077 ± 0.000013	e cos ω	3.7σ (RUWE 35)
A28	CD−32 4901	51.311 ± 0.014	51.37333 ± 0.00020	e cos ω; period	5.7–6.1σ; period 4.5σ
A29	BD−01 648	29.5888 ± 0.0053 (SB2)	29.60666 ± 0.00002	period	3.4σ
A30	BD+29 2845		27.94676 ± 0.00001	e cos ω	5.3–5.6σ
A52	TYC 5406-1905-1	40.543 ± 0.015 (SB2)	40.46953 ± 0.00005	period	4.9σ
A54	TYC 447-247-1	39 (SB2)	3.02348 ± 0.00001	period (spurious)	not an alias; would need 14 + 13.5 $M_\odot$
A63	TYC 3801-1499-1		57.2400 ± 0.0003	e cos ω	3.8σ (4.1–4.3σ from the eclipse spacing)
A65	BD+27 4098		26.768321 ± 0.000009	e cos ω	4.0–5.0σ in the eclipse spacing
A86	BD−01 2238		15.798442 ± 0.000003	period	4.1σ short (eclipse spacing 2.7σ)
A102	HD 194269		1.9385707 ± 0.0000005	period	5.0σ short
A110	TYC 7212-2039-1	22.054690 (SB2)	22.039984 ± 0.000046	period	4.5σ long, 21 min per orbit; the secondary at phase 0.506 also differs from the 0.526 that Gaia's e, ω predict

Label	Name	Gaia P (d)	Eclipse P (d)	Fails in	Discrepancy
A113	TYC 4251-1144-1	37.6646 (SB2, circular)	23.63714 ± 0.00001	period	Gaia's period is the 63-d scan-cadence alias of the eclipse period (1/23.637 – 1/37.665 = 1/63.5 d); two primaries 23.637 d apart within one sector exclude it by themselves, and $e \cos \omega = 0.280$ excludes the circular solution
A117	TYC 629-1389-1	36.6007 ± 0.0172	36.54015 ± 0.00003	period	3.5σ long, 87 min per orbit; Gaia's e and ω hold
A145	TYC 3344-341-1	4.746792 ± 0.000199	4.746808 ± 0.000003	$e \cos \omega$	the secondary eclipse at phase 0.49996 ± 0.00018 gives $e \cos \omega = 0.0000 \pm 0.0003$; Gaia's +0.036 ± 0.008 (e = 0.039) predicts 0.5231 ± 0.0051, 4.5σ away; the orbit is circular and Gaia's eccentricity spurious
A147	TYC 2136-2756-1	14.801518 ± 0.000644 (SB2)	14.804287 ± 0.000006	period	4.3σ short, 4 min per orbit; at Gaia's own epoch the eclipse is 1.06 h (2.4σ) after the predicted conjunction B, and the batch's +0.019 phase offset at TESS epochs was this period error extrapolated
A159	TYC 6635-619-1	11.874084 ± 0.001091	11.870552 ± 0.000001	$e \cos \omega$	the secondary eclipse at phase 0.62842 ± 0.00004 gives $e \cos \omega = 0.2005 \pm 0.0002$; Gaia's +0.252 ± 0.013 (e = 0.299, ω = 32.6°) predicts 0.6603 ± 0.0081, 3.9σ away with Gaia's covariance and 3.3σ without; the period is 3.2σ long as well; Gaia's goodness of fit of 2.6 suggests underestimated errors, and the K dwarf's 15–25% of the light can bias a single-lined solution
A167	TYC 8994-1423-1	4.173881 ± 0.000092 (SB2)	4.1731519 ± 0.0000014	period	8.0σ long, 1.05 min per orbit; at Gaia's own epoch the two eclipses fall 2.9 h and 1.2 h after the predicted conjunctions A and B (0.01–0.04 in phase), so the orbit is the right one and the error is understated: a perturbed double-lined solution (goodness of fit 12.2, RUWE 2.70) on a star with a ±3 min eclipse-timing variation common to both eclipses, which suggests a third body; the secondary at phase 0.49985 ± 0.00012 also gives $e \cos \omega = -0.0002 \pm 0.0002$ against Gaia's +0.0256 ± 0.0097 (2.7σ)
A202	TYC 5995-1398-1	2.5540664 ± 0.0000374	2.553938 ± 0.000005	$e \cos \omega$; period	3.5σ in $e \cos \omega$: the occultation at phase 0.49993 ± 0.00045 gives −0.0001 ± 0.0007 against the +0.051 ± 0.015 of Gaia's e = 0.054 ± 0.015, so the orbit is circular and the eccentricity spurious; and the period is 3.4σ long, 11 s per orbit, with Gaia's conjunction A drifting from +0.1σ to +2.4σ across the TESS sectors; conditioned on a circular orbit, both conjunctions match the eclipses at Gaia's own epoch (−2.0σ and −1.6σ), so the attribution holds; the primary is a spotted RS CVn-like subgiant whose eclipse timings jitter by ±6 min

Discrepancies are in units of Gaia's quoted error. A blank Gaia period is not quoted in the text (for A30 and A65 it agrees with the eclipse period); the Supplement entries give it. Eclipse periods are those of Table A1, except for A29, where the value used in the comparison is given (Table A1: 29.606659 ± 0.000010 d). Three conjunction offsets are in tension without being counted, because they move with the treatment of Gaia's covariance: A87, 4.3σ at Gaia's epoch and 1.7σ at the TESS epoch carried with Gaia's period; A89, 6.7σ with the full covariance of a poorly fitting double-lined solution and 2.2σ with the correlations ignored; A104, 8.2σ with Gaia's epoch carried with the TESS period and 0.2σ with Gaia's own period (its eclipse spacing gives $e \cos \omega$ 2.9σ from Gaia's).

Table 5. Catalogue corrections: the eleven numbered corrections B1–B11, the further corrections from the candidate checks, and the VSX type revisions among the new eclipsing systems.

Label	TIC	Name	Catalogue entry	TESS	Note
B1	48875187	Gaia DR3 5451130921097245056	VSX RS (Gaia classifier), no period	detached twin EB, P 0.862264 d, equal 9.1% eclipses	held: host not separated from two possible neighbours
B2	382523691	ASASSN-V J005242.03-513706.0	VSX ROT 5.3637 d	EB at Gaia's SB1 period, P 4.915053 d, 24 + 12 ppt	
B3	397233697	Gaia DR3 5223177460339438 08	VSX ROT (Gaia); Gao ROT 8.40 d	EB, P 17.13557 d, 100 + 5 ppt	secondary at φ 0.3
B4	333792991	HAT-177-0002605	VSX VAR 9.69 d; ASAS-SN ROT:	EB, P 6.21467 d, 45 + 13 ppt	
B5	385557214	V1274 Tau	GCVS RS 6.62 d; VSX RS 6.448 d; ASAS-3 ESD/EC 1.727 d	EB at Gaia's P 5.62323 d, 4.5 + 2 ppt	ASAS-3 photometry shows only the 6.45-d rotation
B6	464315748	HD 171459	VSX DSCT|GDOR 3.19721 d	EB, P 6.3955 d, 12.7/13.4 ppt	3.19721 d is the eclipse spacing
B7	189269584	CD–28 7251	VSX VAR 0.0146 d	twin EB, P 6.9610 d, 20.7/23.9%	proposed: contamination ratio 1.77 leaves a blend possible
B8	99028319	LP 852-5	TESS-EB (Prša et al. 2022) P 9.4045943 ± 0.00045 d; not in VSX	EB, P 9.400436 d, 32.9/18.5 ppt	
B9	80158129	CzeV3749	VSX EA, no period	P 10.898518 d, no secondary	blend, 9″ from the peak

Label	TIC	Name	Catalogue entry	TESS	Note
B10	120660110	WISE J105540.2-521127	VSX EW/EA 0.8152227 d	*P* 0.817052 d	blend, 11″
B11	306658481	NSV 3445	VSX suspected variable, no type or period	eclipsing, 1.66 d	1.5″; depth sector-dependent
Further corrections from the candidate checks					
	435855781	TYC 1194-680-1	SuperWASP EA/EB 1.66 d (Thiemann et al. 2021); not in VSX	EB, *P* 22.450574 ± 0.000054 d, 45% and 39%	Gaia SB2 22.44398 d, 2.0σ short
		Gaia DR3 3327090079543787 52	VSX E 3.23146 d	EB, *P* 5.725588 ± 0.000018 d, circular	G 14.8, 63″ from TIC 206952704
		Gaia DR3 5956391969014516736	VSX E 12.3202 d	EB, *P* 1.5432 d	G 13.9, 53″ from TIC 34344890
VSX type revisions among the new eclipsing systems (Table A1)					
A20	235666667	HD 170829 = NSV 24480	VSX VAR:, no type or period	EB, *P* 26.384493 d	
A25	258780392	BD+12 121	VSX MISC 91.5 d	EB, *P* 43.8230 d	91.5 d is spot modulation of the giant
A31	2048935928	V1192 Cas	VSX SRA 37.4 d	EB, *P* 37.4133 d	37.4 d is the orbital period
A32	399758284	TYC 97-228-1	VSX ROT 39.81 d	EB, *P* 40.51032 d	39.81 d is the giant's near-synchronous rotation
A52	307170556	TYC 5406-1905-1	VSX MISC 79.3 d	EB, *P* 40.46953 d	

Depths are as measured in TESS (B1 in TGLC photometry). B1 and B7 are proposed and await the further step in their notes; the other nine are established by the data. For B9 and B10 the TIC star is the aperture in which the blended signal was found. The two Gaia DR3 stars of the second block are neighbours of conjunction-window targets, detected through their contamination of the targets' apertures. TYC 1194-680-1 is a confirmed eclipsing binary but not new: the revision is of its period and type. A21's only prior record is an ExoFOP CTOI that is the star's own secondary eclipse.

Table 6. Other objects: the planet candidates C1–C3, variables that are not eclipsing binaries, instructive false positives and rejected candidates, one line each; the Supplement has the details.

Object	TIC	What	Status
Planet candidates			
C1: TIC 4206066 b, c	4206066	two transiting-planet candidates (Rabtsevich 2026), *P* 3.182785 and 11.132690 d, recovered blind by the M-dwarf search	candidates; independently confirmed in public data; not our discovery
C2: TOI-5394.01 = HD 89277	61109252	the TOI signal is the occultation of an M-dwarf companion; *P* 15.195017 d is Gaia's SB1 period	explained
C3: TOI-6284, TOI-1467	286763141, 240968774	three transit signals around known hosts: 7.349277 and 5.244544 d (TOI-6284), 3.154127 d (TOI-1467)	independently recovered; in a preprint
Other objects			
GJ 4348	206327797	0.223864-d double wave; a detached near-twin M+M pair plausibly on top of a young star	open; one spectrum at quadrature settles it
TYC 3763-577-1	322347036	δ Scuti star, 15.32, 20.53 and 27.14 d^{-1}, ~1 ppt (diluted)	uncatalogued
TIC 234495456	234495456	BY Draconis spot rotator at ~0.7623 d	uncatalogued
TIC 189882802	189882802	a 3,444-ppm, 4-h single event made by two flares	dead
TIC 96202086	96202086	0.63048 d, 307 ± 32 ppm; ~40 such peaks expected over 1,299 stars	not a detection
HD 151770	337393383	real 16.3 + 11.2 ppt eclipses; $e \cos \omega$ ~7σ from Gaia's SB2 and both conjunctions miss	open
HD 34757, HD 44342, HD 196402	309416862, 38022998, 100614055	demoted fold candidates of 0.9–1.6 ppt	dead or unresolvable
EBLM J0021–16 = BD–16 52	406352710	published EB recovered blind by the *G* 9–10 fold; *P* 5.9672641 ± 0.0000012 d	known; a blind recovery
BD+30 9	22503559	ellipsoidal variable on a Gaia SB1 orbit, *P* 1.5786673 ± 0.0000010 d, 16.1 ppt peak to peak; no eclipse	new ellipsoidal variable
TYC 3707-251-1	316045876	4%, ~3-h dips every 5.854343 d; withdrawn as an eclipser on a 197-d Gaia astrometric orbit	new 5.854-d EB; triple probable, host not pixel-verified
TIC 271520202	271520202	two eclipses (18.5% and 17.3%) 4 × 192.391 d apart; Gaia 193.38 ± 0.81 d	plausible; one more primary would confirm
TIC 4382254	4382254	a lone 7,328-ppm, 0.72-h dip in sector 70	not a candidate
Marginal folds		eclipse-like folds in no catalogue	recorded, not chased

C3: all three signals are in the preprint of Tschudi (arXiv:2607.23781). TIC 316045876 and TIC 271520202 come from the edge-on astrometric sample (Section 2.6). The TIC numbers of the marginal folds and the Gaia binary suspects among bright late-M dwarfs are in the Supplement.

4. Discussion

4.1 Why the appointments method works, and its yield

Gaia's elements fix the conjunction times in advance: a single dip either lands in a window fixed beforehand or it does not. The search found 14 new eclipsing systems among 1,422 stars at G < 10 (1.0 per hundred) and 49, five of them probable, among 2,327 at G 10–11 (2.1 per hundred): 1.7 per hundred overall. At G 11–11.5 the first twenty of fifty candidates re-analysed were all eclipsing systems, 0.8 per hundred so far among 2,608 stars, with thirty candidates outstanding. At G 11–11.5 the first twenty of fifty candidates re-analysed were all eclipsing systems, 0.8 per hundred so far among 2,608 stars, with thirty candidates outstanding. At G 11–11.5 the first twenty of fifty candidates re-analysed were all eclipsing systems, 0.8 per hundred so far among 2,608 stars, with thirty candidates outstanding. The

period fold of Gaia orbits at P < 20 d gave 0.4 per hundred at G < 9 and 1.3 at G 9–10; at G 10–11.5 the batch fold gave 1.8 candidates per hundred, and 58 of the 60 shallowest candidates, and all eight of the difference-image-flagged ones, survived the per-object re-analysis. The searches were built differently and the rates are not directly comparable.

4.2 What the eclipses say about the Gaia orbits

Eclipses give the inclination and the relative radii that a spectroscopic orbit leaves free. Most systems with a Gaia orbit at the eclipse period agree with it within about 2σ. Nineteen of the 184 counted systems compared with a non-circular Gaia orbit do not, by 3σ or more in period or $e \cos \omega$ under every treatment of Gaia's covariance, and A113's circular solution carries the wrong period (Section 3.4). Among the G 9–10 systems, 13 of 31 non-circular orbits have a largest offset of 2.3σ or more where about two are expected, and Gaia's quoted errors for these orbits appear to be underestimated by a factor of about 1.5. Below $e \approx 0.05$ the eclipses also test Gaia's $e \sin \omega$, through the durations and the depth of the secondary eclipse, and the result is mixed: six orbits want a circular orbit where Gaia found a small eccentricity (A140, A145, A161, A167, A202, A203; the direction of the bias of Lucy and Sweeney 1971), two want more eccentricity than Gaia found (A137, A171; A204 leans the same way at 1.9σ), and one shows Gaia's 2.3σ eccentricity to be real at 19σ (A176). The period, the conjunction at Gaia's epoch and $e \cos \omega$ therefore carry the attribution, and $e \sin \omega$ is treated as a soft prior.

In A54 a 3.02-d eclipsing pair sits under a spurious 39-d Gaia SB2 solution. El-Badry et al. (2026) found 14 hierarchical triples of the same kind, with the eclipse period much shorter than the Gaia period, and report about half of their Gaia SB1 solutions for compact-object candidates to be spurious. The eclipsers here are a different population, but the contradicted solutions show that a Gaia orbit requires independent confirmation.

4.3 The eclipses on the Harvard plates and in ASAS-SN

The ephemerides can also be run backwards. The Harvard plates, digitised by DASCH (Grindlay et al. 2012; data release 7), cover most of the sky from 1890 to 1990 with a scatter of about 0.13 mag per plate at G 10–11, enough to show a 10–40% eclipse if enough plates fall inside the window. The 31 systems among A1–A74 and B1–B11 with an eclipse of 8% or more and G ≤ 12.5 were folded back 600–12,000 cycles, with the cuts, the statistic and a permutation null fixed before the first fold. Seventeen of the 31 show the eclipse at our ephemeris, 15 of them at the TESS period unchanged; A30's 42% eclipse is a 0.6-mag dip on 27 plates (z = 19). Two systems are untestable, six are too shallow for their plates, and six are below threshold with depths consistent with TESS. The last of these, A63, was first read as an absence: nine plates in its window give 0.066 ± 0.036 mag against the 0.21 mag peak depth of its 18% eclipse, and injected eclipses were recovered 10 times in 10. But the injections were boxes at the peak depth and the eclipse is V-shaped, so the window mean it predicts is 0.105 mag, 1.1σ from the measurement; fitted with the TESS eclipse shape, the plates allow 0.62 ± 0.29 of the TESS depth, where nine of the detected systems fitted the same way give a median of 0.95 with a scatter of 0.25 between stars. A63 is consistent with its TESS depth, not absent. The same fold, applied to the five systems among A75–A109 that meet the cuts, adds A108 (TYC 2747-436-1), whose eclipse appears on 258 in-window plates from 1899 to 1990 (z = 19, the third-strongest detection); A83 is suggestive (FAP 0.005) at a period 4 eP below the TESS value; A82 and A86 are not seen, the plates allowing 0.04 ± 0.31 and 0.28 ± 0.23 of their TESS depths; A98 is untestable. Where TESS covers one or two sectors the century of baseline can refine the period: the plates detect A37 at 22.6131 ± 0.0002 d, twenty times tighter than the TESS value, which Table A1 keeps. The century of plates also tests whether the eclipses have changed: for the 19 systems the plates detect, the TESS eclipse profile was fitted for its amplitude in 20-year bins from 1890 to 1990, and no system changed depth at 3σ (three are at 2–2.9σ, where one is expected by chance, and each resolves into a single bad era or a single plate series), so A73 remains the only system whose eclipses have changed. Three systems (A38, A28, A16) are 1.4–2.0 times deeper on the blue-sensitive plates than in the TESS band, a colour term that constrains the temperatures of their components.

ASAS-SN V-band photometry (2014–2018; Shappee et al. 2014; Kochanek et al. 2017) gives a second archival test, independent of the plates and reaching shallower eclipses. Of the 118 objects with an ephemeris, 55 have an ASAS-SN light curve with a measured depth, a duration and an in-eclipse epoch. Folded on the ephemerides of Table A1, a backward extrapolation of 15 to 630 cycles into data that had no part in deriving them, and tested against a null that slides the eclipse window to 10,000 random phases, 20 of the 55 are fainter in eclipse at $p < 0.05$, where 2.75 are expected by chance, and six at $p < 0.01$ (A6, A33, A35, A38, A39, A47), where 0.55 are expected. Seven of the 20 are also on the plates (A4, A6, A33, A38, A47, A55, A56); ASAS-SN recovers 13 that the plate fold could not, four of which (A22, A50, A65, A70) the plates tested and did not detect. For the 11 systems detected under a shuffle-permutation null, measured and predicted depths agree (median ratio 1.04, Pearson $r = 0.785$, permutation $p = 0.003$).

ASAS-3 V photometry (2000–2009; Pojmański 2002) fills the gap between the plates and ASAS-SN for the 49 objects south of declination +28°, 34 of them with usable light curves. Folded on the same ephemerides with the same sliding-window null, nine are fainter in eclipse at $p < 0.01$ (A6, A10, A22, A28, A45, A47, A57, A58, A66), and A54 is a tenth once its window is narrowed to its 2-h eclipse; A47 measures 0.456 mag against 0.451 predicted, 193 cycles back. Five of the ten (A10, A45, A57, A58, A66) have no other non-TESS recovery, and A57's excess depth may be the giant's own variability. Among A75–A109, 13 of the 19 systems with usable curves are detected, out to 4,707 cycles (A106). Calibrating

the sliding-window null on the light curves themselves, with 440,000 fake ephemerides placed away from the eclipses, gives a false-positive rate of 0.006 at $p < 0.01$ and 0.042 at $p < 0.05$, so these counts and the ASAS-SN counts are, if anything, conservative.

4.4 Limits

The completeness of the appointments search was measured by injection: trapezoid primary eclipses at every conjunction of an ephemeris drawn from Gaia's own errors, into the light curves of 300 of the searched stars, run through the search and the pre-set cuts unchanged (7,200 injections). Where TESS covers an eclipse (94% of injections), the cuts pass 79% of eclipses of 5–10% depth, 67–74% at 2%, 50–69% at 1% and 33–56% at 0.5%, the longer durations recovered better; at every depth down to 0.3% the eclipse is in its window at SNR ≥ 8 in 97–99% of covered injections, so the losses are in the cuts. Most of the loss is one defect: the random windows of the null test were allowed to overlap the star's other predicted windows, so a multi-sector eclipser could be scored against its own eclipses and rejected. The reported systems do not depend on it (54 of the 55 pass the cuts with the null corrected, the other was found by eye), but re-running the search with the null corrected passes 343 further stars that the original cuts rejected, which are being examined. The first of them, TYC 7212-2039-1 (TIC 158544874, a Gaia SB2 at 22.05 d), is confirmed and counted as A110: 13–30% eclipses of both kinds in seven sectors, on the star by difference imaging, with an eclipse period of 22.039984 ± 0.000046 d, 21 min per orbit shorter than Gaia's (4.5σ). Eclipses longer than the 14-h maximum box and the false-alarm rate are not measured. Seven of the 54 reported systems were admitted on the remaining pre-set cuts without the z test. The conjunction test inherits Gaia's period: A9 passed it at Gaia's 13.42 d, and its true period is 27.08 d. The literature check (Section 2.7) read the titles of the documents the NASA ADS full-text index returned, so a mention in the body or a table of an unrelated paper, or under a name not queried, could be missed, and it did not include Zenodo, which is how the records of Section 1 were missed. No spectroscopy was used. The sample stops at G 11.

4.5 Predicted eclipses and follow-up

Appendix A gives the next eclipses, in geocentric UTC, for the original 199 A-labelled catalogue-screened systems except A3. Reports for the Appendix C systems give conditional predictions where their timing baseline permits; these should not all be assumed to have minute-level precision. Most predictions are good to a few minutes. Most of the stars are G 9–11, with eclipses 1–40% deep lasting hours, so many are within reach of small telescopes. The TESS pipelines reduce the same pixels; an eclipse seen from the ground at the predicted time is the independent observation. A missed eclipse shows an ephemeris error, and an early or late one measures eclipse-timing variations (hinted at, for example, in A47). Timings of both eclipses measure $e \cos \omega$ and so test the contradicted Gaia orbits; depths of A73 would show whether its fade continues; timings of A95 would test the light-travel time. HIP 1060 (A47), with an annular eclipse on 2026 November 2 at 06:37 UTC, has been submitted to the AAVSO VSX and is pending.

The catalogue is bright (median T = 9.6). Every object saturates ZTF, the 27 with no ASAS-SN source are 0.81 mag brighter in the median than those with one, and ASAS-SN's scatter rises from 0.02 mag at V 11–12 to 0.08–0.09 mag brighter than V 9.5. For about a quarter of these systems a small telescope is the only instrument left that can follow them.

4.6 The null planet search

The transit searches of about 3,100 nearby M dwarfs (Section 2.3) found no new planet. Under the estimator's assumptions, the repeated-transit searches were expected to yield about 1.2 detections, with a probability of none of 0.30 and a range of about 1 to 8 allowed by those assumptions; the single-transit search, which the TOI list does not cover, expected about 3 more and is the only one with a real shortfall (2.5σ). A naive occurrence-rate calculation (Dressing and Charbonneau 2015) gives 75–95. The difference lies in the sample. Every search but the one around known hosts excluded stars with a TOI or CTOI; the cut removed 2.2% of the stars but all 41 known hosts of a detectable 0.5–15 d planet in this population (95% upper bound on the surviving fraction 7.3%). The searches recovered both TIC 4206066 candidates blind and showed TOI-5394.01 to be the occultation of an M-dwarf companion (Supplement, S3.7).

5. Conclusions

1. We identify 470 eclipsing systems in public TESS photometry: 439 confirmed and 31 probable. All meet the specified seven-group uncatalogued standard; fourteen plausible cases remain excluded. Scoped catalogue non-matches do not establish exhaustive literature priority.
2. One hundred seven came from eclipses by appointment. Among 3,749 stars at $G < 11$ it found about 1.7 new eclipsing systems per hundred stars, and a third subsample of 2,608 stars at G 11–11.5 contributes 42 confirmed systems, with five candidates unreviewed. The fresh exploratory appointments pilot adds two confirmed systems; across these populations, 102 systems are confirmed and five probable.
3. Gaia-orbit period folds contribute 355 systems, including the fainter spectroscopic-binary population; four systems come from the nearby-M-dwarf-selected search (including neighbouring hosts), and four from near-edge-on Gaia astrometric orbits. The same physical and catalogue criteria apply across these populations. Known recoveries, plausible signals, non-eclipsing tidal variability and contaminating sources remain visible in the evidence register and are excluded from unique additions.

4. The eclipses contradict twenty Gaia DR3 orbit solutions in period or $e \cos \omega$.
5. Nine eccentric binaries are new heartbeat stars, one of them also eclipsing, and a tenth is a candidate. We also report three probable blends and eleven catalogue corrections.
6. For 17 of 31 deep systems, Harvard plates from 1890–1990 show the eclipse at our ephemeris, and ASAS-SN photometry from 2014–2018 shows it for 20 of 55 systems tested, with the depths matching the predictions in the 11 examined.
7. A transit search of about 3,100 nearby M dwarfs found no new planet where about 1.2 detections were expected under the estimator's assumptions (probability of zero 0.30), because removing known planet candidates removed most stars where a detectable planet is known. Both TIC 4206066 candidates were recovered, and TOI-5394.01 is the occultation of an M-dwarf companion.

HIP 1060 (A47) has been submitted to the AAVSO VSX and is pending; submissions for the others are in preparation.

6. Contributions and AI use

The author posed the question, directed the work, commissioned and read an internal adversarial review of the draft, decided what is claimed and at what status, and takes responsibility for the content of this paper. The searches, measurements, checks and text were carried out by AI language-model agents using local and cloud execution: Claude Opus 5.5, Claude Opus 5, Claude Fable 5 and Claude Fable 5.1 (Anthropic 2026), and GPT-5.6, GPT-6 and GPT-6.1 (OpenAI 2026). In outline, Claude Opus 5.5 designed and ran the M-dwarf transit searches and the first Gaia-orbit fold, explained TOI-5394.01, and carried out the independent re-analysis of every candidate of the appointments search, the G 9–10 fold and the edge-on search, and the review of the G 10–11.5 candidates; Claude Opus 5 ran the late-M searches, the expected-yield analysis of Section 4.6, the live catalogue check and the ASAS-SN archival fold of Section 4.3; Claude Fable 5 ran the scaled Gaia-orbit fold and its continuation to G 11.5; Claude Fable 5.1 devised and ran the appointments method, folded the Harvard plates and assembled the paper; GPT-5.6 ran the edge-on astrometric-orbit search and the look-elsewhere test for TIC 4206066; GPT-6 (Astra) supervised the 174-object follow-up campaign, reviewed exceptional cases, reconciled the evidence and integrated the results; fresh GPT-6.1 Sol leads and GPT-6.1 Terra helpers carried out the detailed photometry, covariance, catalogue and attribution checks. Fable 5.1 led detailed period-fold reviews with Opus 5.5 helpers, preserving report-specific caveats and separating physical evidence from catalogue novelty. These candidates were originally found by Fable 5's scaled Gaia-orbit search. The project coordinator thanks all contributors who assisted with screening, evidence collection, analysis, review and reconciliation. Per-object credits are recorded in the Supplement and repository review register (Appendices B and C).

The adversarial review was run by a separate group of agents with its own code and no access to the agents writing the paper. It re-derived about 2,500 of the paper's numbers from the analysis files and re-reduced eight systems from the raw light curves; a second review of the same kind read the 35 systems added afterwards and re-measured eight of them. Both were internal quality control with independently written code, not journal peer review. Only public data were used; no telescope time, spectroscopy or proprietary data were involved. The use of AI systems is disclosed here and in the Acknowledgements, as the Journal's editorial policy asks.

The completed fainter SB1 review used Sol/Terra measurements under Astra supervision, following original Fable discovery and screening. The project coordinator coordinated the work; agent measurements are attributed to their performing systems.

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The analysis, the checks and the text of this paper were produced by AI language-model agents (Anthropic 2026; OpenAI 2026) directed by the author, as set out in Section 6. We thank P. Rabtsevich, whose public report of the TIC 4206066 candidates prompted the checks of Section 3.6.

The fainter spectroscopic-binary review used Sol/Terra measurements under Astra supervision, following original Fable discovery and screening. The project coordinator coordinated the work; agent measurements are attributed to their performing systems.

We thank all contributors who assisted with screening, evidence collection, analysis, review and reconciliation.

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Appendix A. Ephemerides and next eclipses

Table A1 gives, for the A-labelled confirmed and probable systems, the period and epoch from the eclipses, the eclipse depths and the primary duration where measured, and the next predicted primary eclipse after the date of this paper (none for A3, which has no timing fit), in geocentric UTC (light-travel time to the barycentre removed). The R-labelled and returned C-labelled reviews are registered in Appendix C; their reported ephemerides and limitations are in the review registers and reports. The next secondary eclipses for the A-labelled systems are in the Supplement. Epochs are BJD_{TDB}. The per-object entries in the Supplement give the measurements behind each row.

Table A1. Catalogue of the 199 eclipsing systems (187 confirmed, 12 probable) in label order, with the by-catch variable A46 and the previously typed A with unresolved hosts (A13–A15) in a separate block: ephemerides, depths and the next primary eclipse after 2026 October 14 (after 2026 October 19 fo October 22 for A180–A204).

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (I
A1	311958594	UCAC4 083-054218	9.822	1.2443947 ± 0.0000016[a]	2458629.45849	2.28	1.85	0
A2	183494856	UCAC4 070-000968	14.810	6.129392 ± 0.000006	2457003.74608	21–29	17–23	
A3	85379511	HIP 83929 = G 181-29	9.656	9.474664[b]	2458993.379	0.0515		1.5
A4	70665829	UCAC4 537-050325	12.294	3.4657540 ± 0.0000093	2459528.2147 ± 0.002	30.2	6.73	2.7
A5	22388285	HD 106770	8.982	19.454722 ± 0.000010	2460494.45294 ± 0.00039	2.03		4.3
A6	348773952	BD−13 1046	9.596	5.0782783 ± 0.0000010	2460609.56921 ± 0.00016	10.97	2.79	2.5
A7	271321097	UCAC4 130-138342	10.930	13.9106635 ± 0.0000071	2460685.73790 ± 0.00034	8.00	1.00	3
A8	402800288	HD 131614	8.470	10.4462098 ± 0.0000098	2460391.45705 ± 0.00071	1.82		11
A9	399009076	HD 7601	8.557	27.0822013 ± 0.0000180[f]	2460097.60593 ± 0.00023	39.8	26.1	14
A10	300395201	HD 99010	8.449	7.8127421 ± 0.0000013	2460384.50567 ± 0.0001	2.20	1.73	4
A11	159892657	BD+40 3254	8.418	4.0751904 ± 0.0000009	2460133.82003 ± 0.00011	10.11	1.87	4.0
A12	436758044	HD 217492	8.326	14.0343984 ± 0.0000110	2460203.47040 ± 0.00028	1.82		4
A16	257352286	TYC 2173-965-1	9.423	23.802223 ± 0.000022	2459804.28519 ± 0.00089	10	8.45	2
A17	219230041	HD 26958	7.514	40.45701 ± 0.00002	2457386.08460	1.2		9
A18	189668511	TYC 2819-1027-1	9.216	31.80877 ± 0.00001[g]	2458766.58610	3.75	2.2	
A19	219766411	Gaia DR3 1440245818754478976	12.395	290.9037 ± 0.0006	2458936.3573 ± 0.0028	12.5		
A20	235666667	HD 170829 = HR 6950 = NSV 24480	5.785	26.384493 ± 0.000040	2459403.97980	1.65	0.106	11
A21	403113935	TYC 4244-623-1	9.411	30.763077 ± 0.000013	2459849.89772 ± 0.00041	2.77	0.27	16
A22	324515186	HD 309050	9.444	23.034195 ± 0.000010	2459657.89250 ± 0.00031	8.3	1.1	7
A23	142935468	TYC 4160-760-1	9.454	22.244792 ± 0.000017	2459228.68805 ± 0.00056	1.70	1.8	6
A24	331507968	TYC 3669-262-1	9.278	24.690127 ± 0.000031	2458797.42249 ± 0.00146	0.86	0.32	6
A25	258780392	BD+12 121	8.520	43.8230 ± 0.0025	2459451.28700 ± 0.0013		2.9	16
A26	89031985	TYC 3091-707-1	8.791	42.14749 ± 0.00117	2459645.40345 ± 0.01221	0.29		18
A27	143198335	HD 63923	9.680	41.30385 ± 0.00030	2459963.04837 ± 0.00438		1.07	7
A28	145034339	CD−32 4901	9.251	51.37333 ± 0.00020	2458512.23071 ± 0.002	20–21	5	34
A29	248652521	BD−01 648	9.571	29.606659 ± 0.000010	2459177.83838 ± 0.0005	9.5		5
A30	311297982	BD+29 2845	9.595	27.94676 ± 0.00001	2458994.50441 ± 0.0018	42		10
A31	2048935928	V1192 Cas (= TIC 271237873)	9.791	37.4133 ± 0.0005	2458764.90610 ± 0.018	4.7	0.9	2
A32	399758284	TYC 97-228-1	9.956	40.51032 ± 0.00011[g]	2458463.92360 ± 0.024	1.3–1.4	0.45	
A33	4341778	TYC 7691-3893-1	9.825	45.59795 ± 0.00008[g]	2458573.67953 ± 0.0042	9	2.2	
A34	139285862	CD−45 14957	9.926	45.161458 ± 0.000109	2458354.30334 ± 0.00477	2.3		1
A35	11488700	CD−29 366 (Bok I 16B)	10.299	20.289728 ± 0.000160	2458383.84197 ± 0.00342	3.3		5
A36	43473641	TYC 5326-1527-1	10.344	54.851934 ± 0.000191	2459192.78258 ± 0.0053	1.0	0.39	2
A37	24755724	CPD−25 5943	10.213	22.611537 ± 0.002850	2460793.29985 ± 0.002	18.7	13.4	17
A38	316326405	TYC 3715-870-1	9.933	37.116502 ± 0.000308[g]	2458839.85111 ± 0.00926	8.7	2.5	
A39	22620802	TYC 3396-1373-1	10.210	35.692330 ± 0.000062	2459602.95779 ± 0.0007	2.6	1.25	8
A40	165541068	TYC 986-148-1	10.156	24.7458 ± 0.0025[c]	2460463.59430 ± 0.001	4.6	1.5	13
A41	156182608	TYC 6568-1036-1	10.521	23.53654 ± 0.00017	2458501.94000 ± 0.011		0.7–1.0	
A42	832117928	CD−25 5638B	10.501	22.029385 ± 0.000018	2458508.34974 ± 0.00098	0.5		4
A43	241445833	TYC 4685-2083-1	10.376	23.205492 ± 0.000007	2458389.70430 ± 0.00052	6.9		4.2
A44	355477435	TYC 8461-136-1	10.501	21.106026 ± 0.000006	2458325.49469 ± 0.00051	5.7		3
A45	34344890	TYC 7897-48-1	10.495	31.152794 ± 0.000072	2459382.75960 ± 0.0037	6–7		3
A46	383529972	BD+30 2277 = WDS ES 436	10.118	0.381845 ± 0.000002	2459645.01110 ± 0.0007	4.6	3.4	
A47	70858500	CD−31 59 = HIP 1060	10.005	24.105122 ± 0.000006[g]	2458357.74508 ± 0.00079	36	34	
A48	66900858	TYC 3166-453-1	10.533	43.196065 ± 0.000034	2459804.38481 ± 0.00045	27	10.4	9
A49	286926100	TYC 3870-1689-1	10.010	28.098880 ± 0.000013	2459645.01560 ± 0.00026	2.6	0.45	5
A50	450218810	TYC 3729-162-1	10.396	24.172470 ± 0.000015	2459926.66270 ± 0.00041	11.5	1.6	7
A51	18000700	BD+28 4244	10.192	29.32848 ± 0.00033	2460558.15104 ± 0.060		1.1–1.3	16
A52	307170556	TYC 5406-1905-1	9.821	40.46953 ± 0.00005	2459206.58750 ± 0.0019	6		1
A53	96844936	TYC 8355-1052-1	10.463	36.7755 ± 0.006[d]	2461155.84550 ± 0.0007		1.8–2.4	5
A54	108344775	TYC 447-247-1	10.291	3.02348 ± 0.00001	2460482.03790 ± 0.0001	18.8	11.7	2.7
A55	11156440	BD+60 397	9.529	21.254843 ± 0.000012	2458801.37455 ± 0.00065	19.5	18.1	
A56	367706305	HD 279736	10.354	32.286594 ± 0.000012	2458817.15278 ± 0.00047	17.5	17.2	1
A57	145469097	TYC 7121-1393-1	10.021	43.324287 ± 0.000075[g]	2458530.31198 ± 0.00384	5.8	2.2	
A58	421900559	BD−14 1559	9.563	35.976045 ± 0.000018	2459203.38951	9.8		
A59	372183259	TYC 3187-1132-1	10.394	42.648930 ± 0.000055	2460537.57169	12.3		
A60	142084127	TYC 9189-1130-1	10.022	45.49880 ± 0.00004[g]	2459553.26450 ± 0.00083	0.8	0.3	
A61	468529113	TYC 4308-545-1	9.981	21.552334 ± 0.000007	2459481.89522 ± 0.00022	14.6		4

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A62	464830297	TYC 110-583-1	10.350	22.34089 ± 0.00001[g]	2458440.02660 ± 0.0004	2.1	0.15	
A63	467158573	TYC 3801-1499-1	10.500	57.2400 ± 0.0003	2458868.88650 ± 0.005	18	16	
A64	249895319	BD+50 578	9.770	29.27232 ± 0.00005	2458793.94630 ± 0.0007	1.67	0.15	9
A65	88619181	BD+27 4098	10.018	26.768321 ± 0.000009	2459819.55960 ± 0.00015	10	0.72	7
A66	260563222	CD−41 5677	9.652	38.791921 ± 0.000024	2458553.49853 ± 0.00089	3.0	0.22	
A67	173468180	TYC 3623-443-1	10.296	29.182072 ± 0.000005[g]	2458756.76116 ± 0.00042	3.8	2.3	
A68	80107967	TYC 8136-2645-1	10.580	34.184472 ± 0.000076	2458521.30741 ± 0.00302	1.3		11
A69	377060838	TYC 243-1377-1	10.162	42.463964 ± 0.000016[g]	2458517.53332 ± 0.00059	17	14.6	
A70	70789999	TYC 4094-1571-1	9.761	22.735028 ± 0.000008	2458823.53837 ± 0.00039	8.2	5.1	
A71	87303050	TYC 3747-2187-1	10.200	24.445338 ± 0.000011	2458833.11980 ± 0.00047	9.7	1.4	4
A72	279636729	Gaia DR3 4720373709202359808	12.008	215.0116 ± 0.0002	2458341.4321 ± 0.0009	38	24.2	10
A73	315721826	TYC 4590-364-1	10.808	17.538[e]	2459940.800	33 to 13	14.6 to 6.3	7.0 to 5
A74	427345302		12.10	173.8755 ± 0.0005	2459884.9860 ± 0.0013	4.62	~5 (ZTF g)	10
A75	202282515	TYC 3255-1824-1	9.655	6.929866 ± 0.000015	2460111.42326 ± 0.00065	6.76	1.55	5.0
A76	202710940	TYC 3256-1533-1	9.480	3.3063718 ± 0.0000018	2460299.60466 ± 0.00020	0.81	0.78	2.2
A77	196475343	BD+44 219	9.116	6.370415 ± 0.000005	2460407.33044 ± 0.00027	0.6	0.37	1
A78	268587594	HD 6217	8.837	9.140474 ± 0.000004	2458764.02178 ± 0.00018	3.65	<0.4	2.3
A79	416277351	BD+64 520A	9.362	6.607651 ± 0.000012	2460149.97660 ± 0.00045	0.454	0.169	2.5
A80	19702288	TYC 1310-1984-1	9.479	17.02161 ± 0.00032	2459516.54908 ± 0.00042	7.98	0.77	7
A81	109061795	CD−24 5530	9.338	13.2271715 ± 0.0000023	2460040.42781 ± 0.00013	2.56	0.189	7.2
A82	317250623	CD−83 126	8.980	19.605278 ± 0.000014	2460060.10199 ± 0.00051	9.8	2.06	5.6
A83	7060775	BD+13 1919	9.163	6.422180 ± 0.000001	2459741.55165 ± 0.00012	10.1	4.59	4.1
A84	456339196	TYC 3805-1342-1	9.566	7.1061147 ± 0.0000043	2459530.89350 ± 0.00012	7.86	4.13	8.6
A85	38388435	CD−27 6336	9.141	2.6164745 ± 0.0000005	2460518.67725 ± 0.00008	5.5	0.28	3
A86	170891257	BD−01 2238	9.414	15.798442 ± 0.000003	2459811.62273 ± 0.00024	12	1.4	3
A87	463896609	HD 89725	8.885	2.2366733 ± 0.0000008	2460195.87831 ± 0.00025	1.64	0.36	4
A88	302351416	TYC 3827-170-1	9.422	13.9574485 ± 0.0000066	2459928.45698 ± 0.00023	3.57	<0.4	4.5
A89	388287047	CPD−85 289	9.317	17.638544 ± 0.000019	2460032.65988 ± 0.00069	1.61	<0.2	3
A90	454747287	CD−73 685	9.346	5.6347635 ± 0.0000015	2459691.10727 ± 0.00013	1.17	0.049	4.3
A91	311635507	HD 311112	9.375	18.5428744 ± 0.0000049	2460479.18152 ± 0.00017	6.67	0.87	7.2
A92	148877735	BD+69 653	9.325	3.6074499 ± 0.0000008	2460110.82736 ± 0.00008	1.32	1.32	3.0
A93	175734936	HD 112601	8.955	4.337953 ± 0.000001	2460380.72231 ± 0.00018	0.99	0.93	2.5
A94	245194662	HD 114997	8.917	17.2795467 ± 0.0000052	2460353.15051 ± 0.00022	2.8	0.25	4
A95	154346187	HD 234179	8.984	3.549541 ± 0.000002	2459489.65835 ± 0.00031	0.559	0.509	2.2
A96	426877400	HD 140021	9.623	2.7560089 ± 0.0000006	2460574.15160 ± 0.00014	0.76	0.031	4
A97	351595900	TYC 1529-387-1	9.316	6.647424 ± 0.000005	2459826.14200 ± 0.00052	0.59	<0.2	4
A98	157469799	BD+45 2574 A	8.948	3.5063244 ± 0.0000002	2460172.71372 ± 0.00002	24.5	19.6	3.8
A99	352236055	TYC 3925-1852-1	9.283	18.821610 ± 0.000008	2460160.83034 ± 0.00023	1.49	<0.13	4
A100	425304484	HD 353655	9.464	16.260285 ± 0.000024	2460531.06071 ± 0.00039	3.6	1.3	12
A101	14596588	HD 192721	8.889	4.478718 ± 0.000065	2460818.15412 ± 0.00013	1.44	0.277	2.9
A102	72160748	HD 194269	8.652	1.9385707 ± 0.0000005	2460815.72949 ± 0.00012	6.95	0.71	2
A103	188258837	BD+41 3794	8.916	5.9855010 ± 0.0000159	2460144.94717 ± 0.00108	3.62	3.18	11
A104	353453805	BD+42 3880	9.503	14.3509487 ± 0.0000051	2460309.02608 ± 0.00010	6.09	1.46	8.6
A105	321718006	BD+67 1313	8.837	3.9253291 ± 0.0000007	2460144.07847 ± 0.00008	1.18	0.09	5
A106	352935659	HD 205273	8.707	1.3854241 ± 0.0000004	2460402.27646 ± 0.00009	3.06	2.92	1.9
A107	397680054	BD+32 4452	9.490	2.976903 ± 0.000001	2459681.70466 ± 0.00012	5.57	5.52	3
A108	239649487	TYC 2747-436-1	9.131	6.4161826 ± 0.0000024	2460233.51625 ± 0.00014	21.4	2.8	12
A109	274331697	BD+43 1585	9.616	1.0394878 ± 0.0000005	2460134.11958 ± 0.00009	1.26	0.15	1
A110	158544874	TYC 7212-2039-1	9.902	22.039984 ± 0.000046	2460010.4835 ± 0.004	24–35	21–29	7
A111	20379300	CD-28 8698	9.926	32.60569 ± 0.00003	2460759.1644 ± 0.0012	4.9–5.1	0.50–0.62	7
A112	305944592	TYC 9053-432-1	10.036	34.48723 ± 0.00005	2458626.1182 ± 0.0026	0.86		18
A113	352914772	TYC 4251-1144-1	9.992	23.63714 ± 0.00001	2458722.60576 ± 0.007	14.5–15.4	1.8–2.2	8
A114	65660608	TYC 6528-895-1	10.386	41.56002 ± 0.00006	2459230.72319 ± 0.0010	9.3–9.5		4
A115	447307865	TYC 9038-1390-1	9.788	40.01697 ± 0.00002	2458628.6578 ± 0.0009	1.3		5

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A116	455491894	TYC 3608-1307-1	10.138	57.3498 ± 0.0100	2460564.19008 ± 0.0004	2.3		5
A117	404715740	TYC 629-1389-1	9.485	36.54015 ± 0.00003	2459492.88004 ± 0.00038	21.5	3.3	6
A118	64696335	TYC 6536-1088-1	10.388	41.86524 ± 0.00087	2459238.650 ± 0.021	0.56	0.24	3
A119	280129511	TYC 4320-529-1	10.506	6.348792 ± 0.000015	2461336.598 ± 0.010	0.34	0.28	3
A120	382605794	TYC 8982-3032-1	10.138	2.140795 ± 0.000001	2460055.59400 ± 0.0007	0.31		1
A121	384249385	TYC 9010-4187-1	10.877	7.805870 ± 0.000011	2460653.55840 ± 0.0014	0.33		3.2
A122	384525203	TYC 3612-1132-1	10.470	4.055307 ± 0.000006	2459826.03683 ± 0.00076	0.58	0.34	3.2
A123	78448518	TYC 5962-1229-1	10.575	11.874406 ± 0.000027	2458477.12450 ± 0.00296	0.39	0.18	2.9
A124	310162414	TYC 3371-625-1	10.342	4.551256 ± 0.000007	2458819.32778 ± 0.001	0.41	0.41	2.4
A125	431278074	TYC 3634-2713-1	10.199	6.386066 ± 0.000003	2458744.40878 ± 0.0006	0.53	0.13	3
A126	279648140	CD−61 607	9.903	3.3502765 ± 0.0000006	2459829.361644 ± 0.000167	0.60	0.071	3.4

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (l
A127	74180670	TYC 8737-1439-1	10.569	18.4215088 ± 0.0000285	2460686.04707 ± 0.00105	0.54	0.33	3
A128	219543346	TYC 1643-218-1	10.965	19.205350 ± 0.000065	2459919.47988 ± 0.00199	0.69		4.5
A129	4965877	TYC 7166-2609-1	11.070	10.220874 ± 0.000008	2459901.40080 ± 0.00075	0.68		2.5
A130	240680969	TYC 3261-1420-1	10.173	18.047704 ± 0.000085	2458778.6842 ± 0.0063	0.67–0.85		7
A131	52855832	TYC 4030-58-1	11.048	2.4113702 ± 0.0000015	2459632.73471 ± 0.00044	0.91	0.50	1.9
A132	283862049	TYC 2263-89-1	10.694	10.144315 ± 0.000004	2459997.87878 ± 0.00029	0.80		1.9
A133	359299569	TYC 8666-1738-1	10.666	7.149090 ± 0.000002	2460352.39715 ± 0.00031	0.79		7.0
A134	329314665	TYC 9000-1011-1	11.017	3.7733403 ± 0.0000019	2460773.739862 ± 0.000333	0.94	0.33	2.3
A135	182517223	TYC 9347-571-1	10.362	2.2076198 ± 0.0000003	2460115.008748 ± 0.000100	0.97	0.12	1.3
A136	344947332	TYC 4004-778-1	11.006	10.451460 ± 0.000014	2459954.44026 ± 0.00080	0.90	0.055	7.5
A137	1260529	TYC 6579-2222-1	10.775	10.992454 ± 0.000008	2459566.15999 ± 0.00035	1.09	0.040	3.7

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A138	306477840	TYC 5361-543-1	11.045	5.520586 ± 0.000003	2459474.27805 ± 0.00051	1.02		3.7
A139	1695417	BD−13 3646	9.640	17.797939 ± 0.000010	2459326.38398 ± 0.0002	0.93		4.4
A140	430987029	TYC 2741-1694-1	11.162	9.920766 ± 0.000016	2460364.52570 ± 0.0009	0.86		6.9
A141	116246364	TYC 3093-1294-1	10.918	14.748239 ± 0.000011	2459765.23338 ± 0.00040	1.14		3.3
A142	98952099	TYC 2971-946-1	11.057	5.955676 ± 0.000004	2459444.81721 ± 0.00034	1.11	0.065	3.7
A143	192973862	CD−24 8975	10.335	19.211924 ± 0.000008	2460144.30777 ± 0.00027	1.09		3.4
A144	200398115	TYC 3624-860-1	10.765	7.305664 ± 0.000049	2460170.06233 ± 0.00263	1.20	0.21	8
A145	260789413	TYC 3344-341-1	10.222	4.746808 ± 0.000003	2460126.22496 ± 0.00027	1.43	0.49	3.3
A146	53256660	TYC 4952-149-1	10.680	5.160140 ± 0.000004	2460411.01065 ± 0.00041	1.29		2.2

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A147	403936502	TYC 2136-2756-1	9.875	14.804287 ± 0.000006	2460043.96544 ± 0.00021	1.40		2.8
A148	85944269	TYC 2669-4733-1	10.995	3.021219 ± 0.000003	2459951.66557 ± 0.00045	1.73	1.11	4
A149	93103821	BD−17 691	10.603	1.6350307 ± 0.0000004	2458868.90788 ± 0.00009	1.55	0.58	1.4
A150	7795044	CD−35 10394	9.549	9.4320556 ± 0.0000068	2460194.85288 ± 0.00053	1.50		4.6
A151	324273420	TYC 8977-7461-1	10.571	3.9158631 ± 0.0000015	2460200.00630 ± 0.00037	1.5–1.6	1.5	3.4
A152	123378551	TYC 7652-366-1	10.739	11.010611 ± 0.000006	2459972.09636 ± 0.00034	2.3		2

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A153	375435321	TYC 8187-1030-1	10.892	18.214018 ± 0.000011	2460498.99671 ± 0.00056	2.15	0.13	9.4
A154	375302184	TYC 9075-1173-1	10.921	2.5455862 ± 0.0000005	2460571.13574 ± 0.00015	2.85	0.19	3.3
A155	285385215	TYC 2327-1770-1	10.585	8.2074497 ± 0.0000015	2458797.97260 ± 0.00021	6.5	0.66	2.9
A156	159836610	TYC 7485-869-1	10.753	5.7022928 ± 0.0000004	2459618.89923 ± 0.00007	9.0	0.79	4.7
A157	86036265	TYC 1405-695-1	10.597	11.9972226 ± 0.0000022	2459507.95631 ± 0.00069	10.0	8.6	10

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A158	268342242	TYC 8142-243-1	10.616	14.6202005 ± 0.0000021	2459840.37583 ± 0.00012	12.4	3.3	9.
A159	437230270	TYC 6635-619-1	11.063	11.870552 ± 0.000001	2459937.17038 ± 0.00009	13.7	1.2	4
A160	142240367	TYC 7558-731-1	10.649	7.541350 ± 0.000002	2460464.17253 ± 0.00033	1.91	0.42	3.
A161	288160281	TYC 4564-1186-1	10.986	7.063961 ± 0.000002	2459597.74250 ± 0.00016	1.62	0.67	1.9
A162	258860596	TYC 2205-1326-1	11.193	4.959311 ± 0.000003	2460056.15218 ± 0.00022	1.81	0.14	2.8

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A163	55114385	TYC 1337-1803-1	10.160	4.268746 ± 0.000001	2459870.92006 ± 0.00009	1.78	0.24	1.8
A164	200745563	TYC 8062-84-1	10.521	10.998961 ± 0.000002	2460412.11666 ± 0.00017	1.85		5.1
A165	423927429	UCAC4 797-029409	10.986	1.4271182 ± 0.0000001	2459967.97498 ± 0.00004	1.74	0.13	2.0
A166	299650500	CD−23 4400	10.052	18.337086 ± 0.000027	2460427.74767 ± 0.00092	1.94		4
A167	449475961	TYC 8994-1423-1	10.276	4.1731519 ± 0.0000014	2460037.02702 ± 0.00033	1.92	1.93	3

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (I
A168	234880209	HD 259137	10.144	4.270292 ± 0.000001	2459508.56208 ± 0.00018	1.88		2.2
A169	284533163	TYC 4020-206-1	10.999	2.349378 ± 0.000002	2459995.69666 ± 0.00046	1.63	0.12	4.9
A170	88457347	TYC 4114-604-1	10.560	5.4689808 ± 0.0000012	2459730.21937 ± 0.00008	2.06		3.3
A171	191146789	TYC 3248-1904-1	10.984	7.7886371 ± 0.0000029	2459851.29421 ± 0.00015	2.27		3.4

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A172	434263468	TYC 4282-552-1	9.968	8.077758 ± 0.000004	2460199.37690 ± 0.00022	2.39	0.14	8.(
A173	293844315	BD−02 2880	9.886	14.774809 ± 0.000003	2459810.95109 ± 0.00021	2.25		4.3
A174	348784186	TYC 5452-573-1	9.952	4.0868675 ± 0.0000004	2459978.32495 ± 0.00007	2.64	1.16	3.(
A175	269504659	TYC 4674-364-1	10.474	11.391807 ± 0.000015	2459800.01713 ± 0.00049	2.36	0.15	4.4

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (I
A176	55214475	TYC 4104-606-1	10.318	4.8136773 ± 0.0000007	2459786.95100 ± 0.00008	2.60	0.15	1.9
A177	63858295	TYC 1429-565-1	10.839	3.4331770 ± 0.0000009	2459599.88038 ± 0.00012	2.64	0.25	1.8
A178	323007878	TYC 9406-845-1	11.040	4.7034219 ± 0.0000005	2459822.87791 ± 0.00009	2.55	0.93	3.3
A179	384284045	TYC 3616-2395-1	10.023	3.787011 ± 0.000002	2460277.42043 ± 0.00006	2.89	2.56	2.2

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A180	39307147	TYC 2609-823-1	10.830	23.421703 ± 0.000013	2458989.52348 ± 0.00059	24–26	9.8–10.2	6
A181	124856476	TYC 7081-350-1	10.669	21.680622 ± 0.000006	2459210.25550 ± 0.00040	9.6–10.0	1.7–1.8	6
A182	200222789	UCAC4 667-109998	10.864	43.89933 ± 0.00007	2458741.61295 ± 0.00125	13.5–14.6	6.1–7.0	24
A183	146111348	TYC 8201-1042-1	10.539	49.24084 ± 0.00003	2458565.73335 ± 0.00118	12.4–14.2	7.7–8.4	25
A184	220461013	UCAC2 5492112	10.736	50.85317 ± 0.00006	2459760.1247 ± 0.0011	3.8–3.9	1.2–2.3	53

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A185	427657329	TYC 4478-554-1	10.278	47.52876 ± 0.00004	2459574.2519 ± 0.0005	2.4–2.6	1.1–1.6	38
A186	252806416	TYC 3444-504-1	10.802	42.44229 ± 0.00008	2459633.95593 ± 0.00099	2.9–3.0	0.75	13
A187	170312515	UCAC4 240-011387	10.825	52.31098 ± 0.00005	2459232.33454 ± 0.00097	1.8–1.9	0.9	19
A188	176778312	UCAC4 275-003209	11.085	22.359363 ± 0.000023	2459575.79945 ± 0.00127	9.3–9.7	1.2	7

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A189	74965330	TYC 7686-1190-1	10.844	39.199094 ± 0.000029	2459671.06029 ± 0.00105	10.4–11.1	6.6–7.0	19
A190	307361219	TYC 9193-870-1	10.841	23.828151 ± 0.0000013	2459089.30214 ± 0.00005	22	0.43	3
A191	356308234	TYC 3519-1236-1	10.732	42.860164 ± 0.000019	2458689.14215 ± 0.00057	26–27	15.2	10
A192	197484019	TYC 3619-1706-1	10.908	51.88235 ± 0.00021	2459835.1830 ± 0.0021	7.1–7.3	0.3–0.5	30

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A193	160300523	TYC 5939-78-1	10.852	37.312955 ± 0.000007	2458473.96743 ± 0.00035	11.1–11.6	0.9–1.1	5.
A194	309861809	TYC 8925-649-1	10.361	38.386132 ± 0.000039	2460769.65585 ± 0.00099	1.0–1.1		9
A195	233640767	TYC 3481-179-1	11.065	30.305947 ± 0.000008	2458752.76490 ± 0.00029	15.0–15.9	5.5–5.8	13
A196	233606058	TYC 4216-238-1	10.922	21.737275 ± 0.000004	2458691.30913 ± 0.00020	17.7	17.1	10
A197	300202855	TYC 7612-1193-1	11.135	25.762668 ± 0.0000034	2459211.64639 ± 0.00020	6.4–6.9	0.7–0.8	5

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A198	320805993	TYC 8787-921-1	10.646	56.173887 ± 0.000011	2458676.76580 ± 0.00021	22.5–22.7	5.3–5.4	7.5
A199	391925531	TYC 9380-1063-1	10.637	43.977834 ± 0.000008	2459079.03934 ± 0.00018	2.0		6
A200	374560569	TYC 3697-940-1	10.875	13.893272 ± 0.000012	2459892.25831 ± 0.00039	2.79	0.42	5.5
A201	70465753	TYC 1365-707-1	10.434	6.4475682 ± 0.0000015	2459831.15209 ± 0.00009	2.66	0.23	2.1

Label	TIC	Name	T (mag)	P (d)	T_0 (BJD_{TDB})	Depth prim. (%)	Depth sec. (%)	Dur. (h
A202	348536911	TYC 5995-1398-1	10.088	2.553938 ± 0.000005	2459998.52091 ± 0.00035	2.82	0.38	3.3
A203	315849332	TYC 8972-1194-1	10.678	5.4921244 ± 0.0000015	2459753.70951 ± 0.00034	2.92	0.76	3.7
A204	453601725	TYC 8675-1668-1	10.436	5.5666840 ± 0.0000021	2460765.53158 ± 0.00026	3.04	2.94	2.5
Probable eclipsing binaries with unresolved hosts (not counted)								
A13	58553437		10.228	1.078685		0.126		
A14	71124276		10.504	1.123892		0.082	0.022	
A15	261135533	TOI-6255	10.446	14.486682		0.1399		

Note: T: TESS magnitude from the TIC (every row). P and T_0: the ephemeris of the independent re-analysis or timing fit, to its quoted precision; T_0 is the epoch of the primary (deeper) eclipse, $\mathrm{BJD_{TDB}}$ = BTJD + 2457000. Depths and durations are aperture values in the named pipeline's flux (Section 2.8), not corrected for dilution. For A25, A27, A41 and A53 one eclipse kind has been observed; the source tabulates its depth in the secondary column, and its times are those given as the next primary. A51's primary is seen only as an ingress of at least 8.6%. Next primary: geocentric UTC, from $\mathrm{BJD_{TDB}}$ with the light-travel time to the solar-system barycentre in the star's direction removed (up to ±8.3 min) and TT − UTC = 69.184 s; an observer's local correction is at most 21 ms. Most predictions are good to a few minutes; A31, A32, A35–A38, A40 and A41 to 0.02–0.10 d; A1 and A2 to ±0.004 d and A19 to ±0.005 d (formal errors of the discovery ephemerides); A72 to ±0.002 d; A73 to ±0.3 d. None of these includes event-to-event timing scatter or A109's eclipse-timing variations of about 30 min. The Supplement entries give the secondary epochs and the next secondaries. [a] A1: T_0 is the shallower eclipse; the next primary listed is the deeper one. [b] A3: fold period and box epoch; no timing fit, so no prediction. [c] A40: Gaia's period (one event of each kind). [d] A53: the single interval between its two events. [e] A73: mean period of the primaries, which varies (single cycles 17.39–17.85 d); T_0 is the first of two clean, consecutive primaries (BTJD 2940.800 and 2958.651); the eclipses are fading, and duration and depths give the range (Section 3.2). [f] A9: the period first catalogued, 13.4166 d, was a fold alias. [g] The deeper eclipse is at conjunction B of Gaia's orbit; T_0, the depths and the next primary refer to it, and its duration, which differs from the A eclipse's, is in the Supplement. Origin: App, eclipses by appointment (Section 2.5); Fold, Gaia-orbit period fold at $P < 20$ d (Section 2.4); Batch, the same fold at G 10–11.5 run in batch, followed by the per-object re-analysis (Sections 2.4, 3.2); MD, M-dwarf transit searches (Section 2.3); Ast, edge-on Gaia astrometric orbit (Section 2.6). Notes: SB, a known spectroscopic binary before this work (the eclipses are new); G, Gaia DR3 orbit solution contradicted by the eclipses (Table 4); K, in the unvetted machine-learning list of Kostov et al. (2025, table 2); TCE, an unvetted SPOC or TESS-SPOC threshold-crossing event (72 of the first 104); V, a VSX entry of a non-eclipsing type (Table 5); CTOI, an ExoFOP community TOI (its own secondary eclipse); R, a prior mention of the eclipses (Bryant et al. 2023); HP, eclipse detected on the Harvard plates (17 of the 31 examined); H, host component or a neighbour within one TESS pixel not separated; bc, by-catch, found at a period other than that of the Gaia orbit.

Appendix B. Data and code availability

The unified scientific review register is scientific-review-register.csv. It groups all 314 individual R/C/P reviews by scientific outcome and records method, population, physical verdict, unique-count status and preserved evidence links. The following original registers and directories retain compatible identifiers and archival provenance.

All input data are public: TESS light curves, target-pixel files and full-frame-image cutouts from MAST; Gaia DR3 from the ESA Gaia archive; catalogues from VizieR, SIMBAD, ExoFOP, the NASA Exoplanet Archive and the AAVSO VSX; the Harvard plates from DASCH. The analysis code (Python, with NumPy, Astropy, Matplotlib and Requests), the per-object re-analysis reports, the fitted ephemerides behind Table A1, the figure sources, the object table (`tables/objects.csv`, one row per object with the analysis file behind it), the candidate table (`tables/batch_candidates.csv`: the 266 rows of the G 10–11.5 fold, Section 3.2, with each row's status after review and, for the original 68 individual reviews, their counted labels or non-EB outcomes; the 216 R-labelled reviews have separate physical and novelty statuses without new counted labels) and the Supplement are in a public data deposit, available from the author, under CC BY 4.0, with the code also under the MIT licence. The code was written for this work and has not been packaged for reuse; every light curve it reads is re-fetched from MAST. Files in the deposit written before 2026 October 12 use an older labelling in which this paper's A20–A71 are A21–A72 (HIP 1060, A47 here, is their A48); the object table uses this paper's labels throughout. The ASAS-SN archival fold of Section 4.3 (2026 October 14) postdates the deposit version cited above and is added in its next version. An earlier working draft of this paper, whose main text carries the full entries for the catalogue corrections B1–B11, the probable blends A13–A15 and the other objects of Table 6, is in the deposit and in the public repository (github.com/Snowstar38/eclipses-by-appointment), beside the current version, the object table and the candidate table. The report of the internal adversarial review, the response to it and the consistency checks are kept with the paper's sources in the same deposit.

The R-labelled review data are supplied as `tables/reviewed_extension_2026-10-30.csv` and, in the public repository, `reviews/2026-10-30/`: the original 70 per-object reports, compact measurement evidence, report hashes, review credits and unresolved follow-up tests. This paper is a working v1.0 draft and has not been formally submitted. Historical draft files are retained as provenance, not as submitted or accepted versions.

The fainter SB1 period-fold evidence register is `tables/returned_review_2026-10-30.csv` in the paper sources and `returned-review.csv` at the public repository root. C01-C24 are preserved; C25-C72 follow original roster order. Older reports remain under `reviews/c9a-return/`; all 55 new reports, catalogue receipts and dated tier reconciliation are under `reviews/cloud72-completion-2026-10-30/`. The register records controls, known recoveries, exclusions, physical verdicts and next tests.

The dated catalogue reconciliation, coverage matrix and retained query evidence are in `reviews/novelty-reconciliation/README.md`. It applies the original seven-group catalogue standard; the later coverage recovery and R071–R140 checks are retained in `reviews/2026-10-30-next70/`, all included systems have complete scoped catalogue coverage; historical reports remain archived with their original novelty labels.

R141-R174 and their full reports, compact numeric evidence, audit records and catalogue receipts are retained under `reviews/2026-10-30-final34/`. The current R register contains 216 reviews; R001-R140 retain their earlier keys and reports.

The conjunction-selected reconciliation is supplied as `tables/appointments_review_2026-10-31.csv` and public appointments-review.csv; reports, deciding evidence and catalogue receipts are under `reviews/2026-10-31-appointments/`. P001-P023 follow the preserved queue order. Epoch kind, period origin and uncertainty qualifications accompany every row. These provenance keys do not define a separate search method.

The current period-fold and exploratory adjudications are supplied in public gap-review.csv; the dated evidence archive retains the full attempt ledger, reports and catalogue scope. New epochs, error meanings and limitations remain per-object quantities.

Appendix C. Physical classification and review evidence

The physical sample contains 470 systems (439 confirmed and 31 probable). Fourteen plausible signals are excluded. Search methods contribute 107 systems from predicted spectroscopic conjunctions, 355 from Gaia-orbit period folds, four from the nearby-M-dwarf-selected search and four from near-edge-on Gaia astrometric orbits. Physical status, source attribution, Gaia-orbit association and catalogue status remain separate.

The 314 individual reviews contribute 271 unique additions to the original 199-system catalogue. Stable R/C/P identifiers are archival keys, not scientific populations. R175-R214 cover the frozen G 11.5-12 queue; R215-R216 cover the near-orbital-period exploratory cases; P024-P026 cover the fresh conjunction pilot, including the known recovery P024. Historical identifiers and report bytes are preserved. Pilot attempts and unavailable data are retained in the separate attempt ledger, with no unbiased yield claim.

Included confirmed systems (252)

Key	TIC	Population	Verdict	Added	Evidence / limitation
R001	373138363	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Obtain stellar-radius constraints and joint RV/eclipse characterization.
R002	270181578	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Characterize high-RUWE multiplicity with high-resolution imaging and better primary mass/radius; then joint light/RV fit.
R004	362610608	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Compare per-sector apertures to explain s99 depth decrease; improve PRF and stellar characterization.
R005	399501645	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Improve weak secondary shape and stellar-radius/dilution characterization.
R006	295945076	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine secondary timing and Gaia small-e orbit; multi-sector calibrated pixels
R007	372376841	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolve SAP/PDCSAP dilution difference and component radii
R008	416525506	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Inspect S55 cycle232 background-sensitive event; refine component geometry
R009	451493289	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolve catalog radius versus joint eclipse geometry; sector baseline variation
R010	348737250	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine component radius/light ratio and calibrated localization
R011	158545440	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.
R012	339787086	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Jointly refit RV and eclipse timings; assess sector-dependent baseline effects and the separate 281-day astrometric solution.
R013	341432079	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.
R014	144749615	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.
R015	304033253	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.

Key	TIC	Population	Verdict	Added	Evidence / limitation
R016	117919258	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Obtain a second sector to verify the secondary ephemeris and reduce extrapolation uncertainty.
R017	97509220	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Obtain another eclipse sector to improve local period and test the weak secondary against phase controls.
R018	148618714	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.
R019	266540667	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.
R020	285145433	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine component radii, dilution and eccentric orbit with joint RV/eclipse modeling.
R021	240360239	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Obtain joint RV/eclipse and stellar-radius constraints for characterization.
R022	341735257	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Obtain joint RV/eclipse and stellar-radius constraints for characterization.
R023	306553279	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Characterize temperatures/component assignment and test the covered null secondary with resolved multiband photometry.
R024	56534111	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Obtain joint RV/eclipse and stellar-radius constraints for characterization.
R025	31357336	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refit the Gaia small-e orbit jointly with both eclipse ephemerides.
R026	342454977	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Obtain joint RV/eclipse and stellar-radius constraints for characterization.
R027	455214882	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Constrain grazing geometry and component light with resolved multiband photometry.
R028	140477577	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Obtain joint RV/eclipse and stellar-radius constraints for characterization.
R029	190972398	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Obtain joint RV/eclipse and stellar-radius constraints for characterization.
R030	349273287	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Calibrate a multi-source empirical PSF and obtain less diluted eclipse photometry.
R031	273724854	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling
R032	365181558	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling
R033	370251176	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling
R034	119301005	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling
R035	377656542	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling
R036	194609413	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling

Key	TIC	Population	Verdict	Added	Evidence / limitation
R037	323434013	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling
R038	327209313	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling
R039	330659284	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling
R040	82620363	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Resolved multiband eclipse photometry / joint RV+light-curve component modeling
R041	30311062	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.
R042	290592570	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.
R043	257800191	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.
R044	6946549	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.
R045	149021625	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.
R046	362082706	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.
R047	436727157	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Constrain SB2 flux ratio/component radii and test the missing conjunction with resolved photometry.
R049	67651413	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.
R050	244352837	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Refine diluted two-star dimensions and small-e orbital solution with joint light-curve/RV modelling.
R051	93213423	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Retry unavailable catalogs; refine joint orbit and stellar characterization.
R052	369984032	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Retry unavailable catalogs; refine joint orbit and stellar characterization.
R053	345589621	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Retry unavailable catalogs; refine joint orbit and stellar characterization.
R054	135251275	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Retry unavailable catalogs; refine joint orbit and stellar characterization.
R055	435190459	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Retry unavailable catalogs; refine joint orbit and stellar characterization.
R056	150696021	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Retry unavailable catalogs; refine joint orbit and stellar characterization.
R057	345050437	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Retry unavailable catalogs; refine joint orbit and stellar characterization.
R059	415751695	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Retry unavailable catalogs; refine joint orbit and stellar characterization.

Key	TIC	Population	Verdict	Added	Evidence / limitation
R060	335862015	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Retry unavailable catalogs; refine joint orbit and stellar characterization.
R061	211231319	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.
R062	120522352	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.
R063	192198579	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.
R066	157000246	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.
R068	253923749	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.
R069	389527961	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.
R070	196426780	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint diluted light-curve/RV modelling to refine radii, flux ratio and small-e geometry.
R071	382423586	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R072	194733085	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R073	195483879	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R074	161817918	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R075	202277216	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R076	268636467	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R077	397594001	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R078	9885893	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R079	414008363	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R080	61106869	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse characterization; retain Gaia period caveats.
R081	39478038	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R082	381591400	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R083	333072525	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.

Key	TIC	Population	Verdict	Added	Evidence / limitation
R084	89329656	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R085	359480060	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R086	242057470	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R087	298432323	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R088	161478713	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R089	309981203	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R090	301961777	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R091	443053687	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R092	311147906	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R093	81481443	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R094	143517607	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R095	165258931	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R096	431535710	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R097	242199703	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R099	5343763	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R100	256477680	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R101	353848852	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R103	364120441	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R104	100249241	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R105	138180245	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R106	39013443	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.

Key	TIC	Population	Verdict	Added	Evidence / limitation
R108	144169003	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R109	27987072	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R110	326623940	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R111	320195702	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R112	164946466	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R113	424249116	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R114	229238588	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R115	293946483	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R116	140802498	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R117	4374712	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R118	188085904	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R119	45006630	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R120	242656909	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R121	90563464	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R122	115549556	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R123	29741740	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R124	156851362	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R125	273177449	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R126	48829357	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R127	162075402	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R128	177285916	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.

Key	TIC	Population	Verdict	Added	Evidence / limitation
R129	144500102	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R130	417917895	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R131	26804032	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R132	220425369	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R133	311918550	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R134	233269866	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R135	42318308	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R137	140246270	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R138	302274338	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R139	122024194	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R140	287032105	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; See target report for limitations.
R141	289404005	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R142	335001355	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R143	21335382	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R144	119830214	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R145	170081245	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R146	305537899	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R147	757171	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R148	427330878	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R149	416848010	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R150	309600815	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.

Key	TIC	Population	Verdict	Added	Evidence / limitation
R151	431698773	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R152	424878861	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R154	350335402	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R155	48777160	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R156	382260140	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R157	162234490	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R158	358766055	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R159	37307440	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R160	89575652	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R161	266486943	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse modeling and longer-baseline eclipse timing; resolve remaining alias and geometry uncertainty.
R162	129280729	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse modeling and longer-baseline eclipse timing; resolve remaining alias and geometry uncertainty.
R164	167759819	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse modeling and longer-baseline eclipse timing; resolve remaining alias and geometry uncertainty.
R166	100014634	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse modeling and longer-baseline eclipse timing; resolve remaining alias and geometry uncertainty.
R167	259853508	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse modeling and longer-baseline eclipse timing; resolve remaining alias and geometry uncertainty.
R168	302342078	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse modeling and longer-baseline eclipse timing; resolve remaining alias and geometry uncertainty.
R169	28634340	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse modeling and longer-baseline eclipse timing; resolve remaining alias and geometry uncertainty.
R170	178080174	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Joint RV/eclipse modeling and longer-baseline eclipse timing; resolve remaining alias and geometry uncertainty.
R171	11661051	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R172	157948102	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R173	61107057	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.
R174	313274300	G 10-11.5 spectroscopic-orbit population	CONFIRMED	true	Report; Fit joint RV/eclipse geometry and a longer timing baseline; retain same-sector period limitations.

Key	TIC	Population	Verdict	Added	Evidence / limitation
C02	468181994	fainter SB1 period-fold population	CONFIRMED	true	Report; Period -2.31 sigma from Gaia; RUWE 12.6/unresolved companion risk; independent pipelines agree about 2 sigma. Literature cones limited to enumerated catalogues, not proof of novelty.
C03	470235665	fainter SB1 period-fold population	CONFIRMED	true	Report; QLP-only secondary phase drift unconfirmed; 4-5% pipeline scale offset. Enumerated catalogue cones and SIMBAD SB* do not establish exhaustive novelty.
C04	379251709	fainter SB1 period-fold population	CONFIRMED	true	Report; Secondary fold-level, not localized; primary securely on target. QLP/SPOC secondary phases differ 2.6 sigma. Radius tension retained.
C06	331293467	fainter SB1 period-fold population	CONFIRMED	true	Report; Heavy blend; QLP plus independent own TESScut satisfies available pipeline check; depths model-dependent 80-116 ppt, ratios/duration agree. Gaia geometry depends on covariance. Gaia-VEB/varisum/SB9 cones lack successful positive controls; no exhaustive novelty claim.
C07	283532878	fainter SB1 period-fold population	CONFIRMED	true	Report; Use successful s12d recheck, not false-empty first archive pass. Known EB 29.7 arcsec away at P 1.7975 d excluded as source; up to 3 ppt contamination possible. Phase drift tentative.
C08	407877153	fainter SB1 period-fold population	CONFIRMED	true	Report; Secondary pixel signal weak individually but summed delta chi2 >=166; QLP depth diluted. Use successful s12d catalogue rerun; all-catalogue query unreliable.
C09	15855794	fainter SB1 period-fold population	CONFIRMED	true	Report; Exclude flagged E124 from primary pixel stack as report does. Catalogue positive controls succeeded; catalogue absence only within searched scope.
C10	92881948	fainter SB1 period-fold population	CONFIRMED	true	Report; VSX ROT/Gaia SOLAR_LIKE is prior variable classification, not prior EB; catalogue positive controls succeeded. QLP dilution explains shallower depth.
C11	468827621	fainter SB1 period-fold population	CONFIRMED	true	Report; Primary localized in two sectors both camera 2; secondary untested in pixels and neighbour could supply in worst case. Literature search skipped, though catalogue cones performed; known Gaia short-timescale variable is not prior EB.
C12	416285595	fainter SB1 period-fold population	CONFIRMED	true	Report; Returned CONFIRMED with caveats: QLP/SPOC s40 61.7 vs 54.7 ppt (~12%, ~4 sigma); TESScut depth unreliable. Secondary not pixel-tested; geometry weak/covariance-dependent. Catalogue cones performed but literature search skipped. Report itself says strict depth agreement would imply PROBABLE; no new stricter tolerance imposed here.
C19	23112764	fainter SB1 period-fold population	CONFIRMED	true	Report; See target report for deciding limitation and next check.
C20	373062090	fainter SB1 period-fold population	CONFIRMED	true	Report; See target report for deciding limitation and next check.
C21	465073693	fainter SB1 period-fold population	CONFIRMED	true	Report; See target report for deciding limitation and next check.
C22	316405449	fainter SB1 period-fold population	CONFIRMED	true	Report; See target report for deciding limitation and next check.
C24	441741446	fainter SB1 period-fold population	CONFIRMED	true	Report; See target report for deciding limitation and next check.
C25	280075497	fainter SB1 period-fold population	CONFIRMED	true	Report; See target report for deciding limitation and next check.
C26	276514049	fainter SB1 period-fold population	CONFIRMED	true	Report; See target report for deciding limitation and next check.
C30	104736355	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.
C32	450055243	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.
C35	267222866	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.
C36	66315917	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.
C37	93783392	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional spectroscopy or higher spatial resolution photometry would refine characterization; no additional deciding gate is required by this representative review.
C38	470947699	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional improved spectroscopy and photometry would refine binary characterization.
C39	135230723	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional improved spectroscopy and photometry would refine binary characterization.
C40	36083554	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional improved spectroscopy and photometry would refine binary characterization.

Key	TIC	Population	Verdict	Added	Evidence / limitation
C41	447566809	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional improved spectroscopy and photometry would refine binary characterization.
C45	360516818	fainter SB1 period-fold population	CONFIRMED	true	Report; Optional improved spectroscopy and photometry would refine binary characterization.
C56	4646090	fainter SB1 period-fold population	CONFIRMED	true	Report; Characterize dilution and grazing geometry if needed.
P001	262653359	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Period uncertainty widened to 0.00008 d: saved floor-fit output gives 0.000072 d, wider than prose 0.00005. Mass and evolutionary state leave factor ~2 light-ratio uncertainty. Next test: Spectroscopy or colour eclipse photometry to constrain companion light.
P002	39013890	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Unresolved third light remains possible (ipd multi-peak 69%; RUWE 1.55), affecting derived radii and geometry precision. Next test: Resolve close companion or constrain third light spectroscopically.
P003	325399242	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Historical prose claims seven-event P=28.038374 +/-0.000191, but saved ephem_A and Afix each have five rows. Adopt corrected five-event artifact. Capable 13.2 arcsec neighbour is disfavoured at roughly 3–4 sigma per sector with systematic floor, not mathematically impossible. Next test: Additional orientation/source-resolved imaging would strengthen close-neighbour rejection.
P004	384026617	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Historical 0.6 arcsec WDS pair is doubtful but unresolved by TESS; attribution is to the Gaia source footprint. Gaia statistics alone do not prove pair spurious. Next test: High angular resolution imaging to check historical WDS component.
P005	389013422	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Secondary phase-0.535 hint is unconfirmed; do not quote it as a measured eclipse. Absolute physical parameters remain model dependent. Next test: RVs to constrain eccentricity and secondary predictions.
P006	4808224	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Single-sector s89 pixel test; pulsation affects secondary depth/timing. Discard s35 secondary fit: all input points flagged and optimiser wandered. Approximate physics.py formula has pi/4 error; exact Kepler-grid result is used. Next test: Another orientation pixel test or improved RV orbit would reduce association uncertainty.
P007	23999420	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Adopt primary timing uncertainty 0.000044 d rather than joint model formal 0.000015 d. Secondary 9.7-sigma pixel regression is historical numerical evidence; representative image has no colour scale. Next test: Recover missing/flagged s91 primary only if additional refinement needed.
P008	112232265	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Intrinsic depth approximately 30–36 ppt; custom 3x3 is diluted (~19 ppt), QLP background affects depth. RUWE 3.51 allows third-body uncertainty. T16 parse with zero cadences is not evidence. Next test: Spectroscopy and source-resolved eclipse depths to constrain companion.
P009	275065610	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; TASOC depth is diluted and its epoch is about 14 min later than QLP; does not invalidate its eclipse detection. Dark-companion fit neglects a few percent secondary light. Next test: A future raw A eclipse or double-lined spectrum
P010	469755925	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; S94 A is quality-flagged; confirmation also rests on clean s13 and s67 A events. B pixel depth5.93 +/-0.81 ppt exceeds pipeline2.8–4.0 ppt; low-S/N/background-template extraction is attribution only, not adopted depth. Next test: Another clean secondary for depth and geometry
P011	356435413	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; CTOI356435413.01 is a prior planet-candidate signal at matching eclipse period; annotate separately from EB novelty. Original report overstates 1.5arcsec target offset: that is deficit-to-light centroid; s79 coordinates imply ~2arcsec to target, s53 ~3.6arcsec. No full limb-darkened grazing fit. Next test: Grazing-occultation model or spectrum to test light ratio
P012	255077215	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; S65 secondary arrives31–42min late after common-mode correction; unexplained. This does not remove repeated A timing or agreement of earlier B events. Quoted period error is original formal fit and predictions need timing-systematic allowance. Next test: Clean secondary to resolve s65 timing shift
P013	262471565	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; B is undetected: expected570–1100ppm versus ~600ppm central-phase limit creates mild model tension; permissible phase/temperature/impact values remain. Period error includes20min pipeline-scatter floor. Next test: Unflagged B-window photometry at <=300ppm
P014	126758342	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Gaia eccentricity and omega broad; A misses in83% of permitted historical model draws, not certainty. Primary_epoch field uses detected deeper B eclipse. Next test: Refined RV orbit or next B eclipse

Key	TIC	Population	Verdict	Added	Evidence / limitation
P015	178486571	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; S87 egress cut by gap. Three-event near-zero timing chi-square makes formal error optimistic; allow >=5min in predictions. Next test: Clean B coverage or another A event
P016	395790195	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Adopt timing-only B period/error; tighter joint-model55.460455 +/-0.000021d is conditional on model. Only two same-kind epochs, cycle count constrained by Gaia. Next test: Another B eclipse to test cycle count
P017	466336052	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Only two A eclipses; no internal timing residual test. B windows are contaminated or missing; no B detection. Gaia conjunction uncertainty is 3-4 d. Next test: Observe a third A eclipse and a clean B window near 0.3% precision.
P018	64066759	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Long-baseline precision partly depends on a fixed-shape partial s7 event; three events total. Simplified geometry favours smaller primary radius than TIC. No B expected. Next test: Fit a full limb-darkened joint A model and constrain primary radius spectroscopically.
P019	49416839	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; A3 lacks egress and is weak. B-A phase separation is 2.3 sigma from Gaia; simple joint geometry has limited constraints. Dominant deep eclipse is labelled B in source. Next test: Obtain a complete A event and joint free-eccentricity model to test the phase-separation tension.
P021	60583013	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Period relies on two secondaries, one after a data gap; one primary only. Older B pixel test inconclusive. Duration geometry is approximate. Next test: Obtain a second primary and joint limb-darkened model to test phase separation and geometry.
P022	277542587	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; TESS A mostly flagged; only one TESS A+B pair. KELT period precision is heuristic and anchored to TESS; broad ground aperture dilutes depth. Sparse KELT B is supporting only. Historical exact-night count not adopted. Next test: Obtain another full well-calibrated A/B pair and an event-timing based period uncertainty.
P023	12864953	G 11-11.5; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Only two B events and one clean A; A ephemeris assumes B period. New A window is entirely quality 2048 flagged. Phase-separation tension about 3.35 sigma remains. Conservative period error is an estimate, not calibrated coverage. Next test: Obtain a clean repeated A and refined radial-velocity orbit to resolve eccentricity/phase tension.
R175	300217120	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; The nearby other eclipse in S67 contaminates a wider baseline window; the final fit uses a narrower 1.65-search-duration half-window and shows the repeated primary. Historical S27 TESScut contains one event but QLP independently shows three primaries in S67/S94. Next test: Optional precision characterization
R176	424716057	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; QLP primary depths vary across apertures/sectors (205–230 ppt in the selected sectors); no common intrinsic depth is claimed. Next test: Optional precision characterization
R177	265350422	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Three G-band-capable neighbours at 32.85, 42.38 and 51.78 arcsec lie away from the compact deficit on the target. The target-centered dip grows rather than vanishes in the small aperture. No calibrated centroid confidence ellipse is available. Next test: Optional precision characterization
R178	274941915	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Three capable neighbours at 32.98, 40.57 and 47.81 arcsec are spatially separated from the target-centered compact deficit. The small-aperture reduction retains the eclipse. No calibrated centroid confidence ellipse is available. Next test: Optional precision characterization
R180	245474316	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Deep repeated eclipses are isolated in both reductions; all four measured primary events support the timing solution. Next test: Optional precision characterization
R181	397586559	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Unequal alternating eclipses (primary median approximately254 ppt, secondary164 ppt) establish orbital period twice the historical4.93316d event spacing. Only deeper even cycles enter the reported primary ephemeris. Historical S70 difference image mixes both eclipse kinds, so its aperture depth is a mixture. Gaia nominal conjunction is0.28093d later at the TESS epoch; measured-Gaia period difference−0.00126495d accumulates this offset. Projecting toward the Gaia2016 reference reduces offset to−0.0166d; the orbit association is physically consistent with an imperfect Gaia period, and the tension is retained rather than subjected to a sigma veto. Next test: Optional precision characterization
R182	305801065	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; The shallow eclipse is clearly repeated with localized ingress/egress despite higher scatter in newer sectors. Three capable neighbours at28.07,33.30,54.57arcsec are separated from the compact target deficit. Depth depends on dilution and trapezoid shape; historical median3.8ppt is not a precision intrinsic depth. Next test: Optional precision characterization

Key	TIC	Population	Verdict	Added	Evidence / limitation
R183	345794632	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Large historical PSF fractional residual0.57 reflects imperfect scaled out-image morphology in a crowded field. No listed neighbour within60arcsec can supply the approximately73ppt eclipse under the G-band flux-budget screen, and the lost-light core coincides with the target. The weaker localization calibration remains a limitation. Next test: Optional precision characterization
R184	258621299	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Unequal alternating eclipses (primary median approximately180ppt, secondary131ppt) establish orbital period twice the historical3.49299d event spacing. Only deeper even cycles enter the reported primary ephemeris. Historical S20 difference image mixes both eclipse kinds, so its aperture depth is a mixture. Next test: Optional precision characterization
R185	65346146	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Repeated flat-bottom eclipses; sector-dependent dilution changes fitted depths from about46 to38ppt. Next test: Optional precision characterization
R186	416646907	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Odd historical half-period cycles are deeper in each of three sectors (about18 versus16ppt); deeper odd cycles alone give the primary ephemeris and doubled orbital period. Alternative-reduction stack mixes eclipse kinds. Approximate central duration permits a noncentral chord. Next test: Optional precision characterization
R187	401138717	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Three capable neighbours and high scaled-out-image residual0.976 limit PSF precision. Target-centered compact loss and aperture-depth growth support attribution. Known EB42.73arcsec away has P0.363740d, distinct from19.79d. Gaia period differs by0.00556d, accounting for much of the accumulated conjunction offset; this physical tension is retained. Next test: Optional precision characterization
R188	334379409	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Repeated raw eclipses have modest baseline variability. Gaia period mismatch accumulates over the baseline; propagated covariance and conjunction comparison are descriptive rather than a sigma threshold. Next test: Optional precision characterization
R189	345406164	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Deeper even cycles near75ppt alternate with47ppt odd eclipses in sector81, establishing doubled orbital period. Primary-only ephemeris used. Historical stack mixes eclipse kinds. Three capable neighbours are away from compact target deficit. Next test: Optional precision characterization
R190	152400959	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Known EB27.09arcsec away at0.257384d differs from5.513d signal. Two capable neighbours and residual0.959 limit PSF precision; compact target-centered deficit and small-aperture eclipse support attribution. Next test: Optional precision characterization
R191	152086769	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Deep flat-bottom repeated eclipses; sector96 aperture depth is lower than earlier sectors. No capable neighbour within60arcsec in approximate G flux budget. Next test: Optional precision characterization
R192	270260727	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Shallow localized repeated eclipses are resolved in sectors67 and95; raw scatter and trapezoid approximation limit precision. No capable neighbour within60arcsec under G flux budget. Next test: Optional precision characterization
R193	112827464	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Raw local baseline slopes are fitted explicitly. One capable neighbour is separated from the target-centered deficit; no calibrated centroid confidence ellipse. Gaia nominal phase offset0.063 and period drift remain explicit; full covariance and near-reference projection support association without arbitrary sigma veto. Next test: Optional precision characterization
R194	423657976	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Repeated isolated eclipses with stable approximately94ppt QLP depth. No capable neighbour within60arcsec under approximate G flux budget. Next test: Optional precision characterization
R195	343926224	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Five clean primary repeats determine the ephemeris. S76 historical cycle 154 has structured excursions and an anomalously weak fitted dip; S84 cycle 176 has a broad distorted profile. Both remain in raw-windows.csv and plots but are excluded from the timing fit. The capable neighbour at 55.00 arcsec is separated from the compact target deficit. Aperture and local baseline variations limit depth precision. The retained timing fit has reduced chi-square 18.08; residual timing scatter materially inflates the stated period error and shows model/baseline limitations. Next test: Optional precision characterization
R196	53993311	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Aperture-dependent primary depth changes from about 107 ppt in S68 to 85 ppt in S95; the same localized eclipse morphology repeats. Gaia conjunction uncertainty is substantial; a phase offset alone is not a veto. Next test: Optional precision characterization

Key	TIC	Population	Verdict	Added	Evidence / limitation
R197	43785616	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Observed trapezoid duration exceeds a central estimate based on default primary mass and minimum companion mass. That approximate estimate is not an upper bound: stellar properties and inclination are not fitted. Repeated flat-bottomed events and matching period support association; detailed binary modeling remains optional. Next test: Optional precision characterization
R198	397178704	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Unequal alternating primary and secondary eclipses establish twice the historical 5.50484 d event spacing. Only deeper odd cycles enter the primary ephemeris. The historical stack mixes both kinds. Large PSF residual 0.676 reflects surrounding light rather than an off-target core; the one capable neighbour lies 45.82 arcsec away from the compact deficit. Next test: Optional precision characterization
R199	77550184	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Repeated isolated eclipses have aperture-dependent depths of approximately 56 to 63 ppt. No capable neighbour lies within the 60 arcsec screen. Next test: Optional precision characterization
R200	314041837	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; The target is a large star (TIC radius 5.09 solar radii) with ellipsoidal modulation plus localized eclipses. A two-harmonic baseline fitted outside the primary core in S77 leaves a 30.31 ppt localized deficit; its double-wave modulation does not explain the eclipse. See harmonic-test.json and png. The 18.79 h local-linear trapezoid width includes smooth variability and is not a physical eclipse duration. The reported depth likewise describes that local model, not intrinsic eclipse depth. Capable neighbours at 27.40 and 53.17 arcsec are away from the target-centered deficit; PSF residual 0.595 limits morphology precision. Next test: Optional precision characterization
R201	7446763	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Primary eclipses are approximately twice secondary depth. Only deeper odd historical half-period cycles determine the ephemeris; the reported orbital period doubles the historical 2.95595 d spacing. Historical S98 aperture stack mixes both eclipse kinds. Next test: Optional precision characterization
R202	22705959	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; A capable neighbour at 39.75 arcsec is separated from the target deficit. The target sits near a pixel boundary: the global deficit peak rounds to an adjacent pixel, while the centroid is only 3.64 arcsec from the actual target. This is not evidence for a different source. Next test: Optional precision characterization
R203	62940616	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; One capable neighbour at 22.67 arcsec requires a 0.329 fractional eclipse to explain the signal, but the lost-light centroid lies 0.205 pixel from the target versus 1.33 pixels from that neighbour, displaced away from it. Its pixel fractional loss 0.0416 is a blended-pixel value and does not measure the neighbour intrinsic eclipse depth. The target-centered depth increases in the 1x1 aperture, consistent with target dilution in larger apertures. TESS resolution and G-to-TESS colour differences limit attribution precision. Next test: Optional precision characterization
R204	100911870	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Four clean primary events in S56 and S83 support the timing solution and approximately 84 ppt raw QLP depth. No capable neighbour is present in the 60 arcsec screen. Next test: Optional precision characterization
R205	77176486	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; The 13.75369 d signal lies near the TESS orbital timescale, but repeated localized raw SAP dips in S31 and S98 and the target-centered cutout deficit support astrophysical eclipses. Quality-nonzero cadences are excluded; no flagged or uncovered window is a clean null. Cycle188 is visibly present but its trapezoid covariance is singular, so it is excluded from timing. Bounded S98 pixel/background check (background-check.json) finds target-pixel losses11.99/22.43/16.35 electrons/s versus3.24/9.85/7.73 at the close capable neighbour and0.70/6.30/3.36 at the farther one. Outer-pixel median losses are -0.027/5.68/4.04: common-mode background affects later events, while cycle187 is localized with negligible outer deficit. Core quality flags are0/0/3; flagged cadences are excluded. FLUX_BKG is zero throughout and supplies no independent background measurement. These are descriptive diagnostics, not a calibrated artifact veto. Gaia conjunction offset is 0.04208 phase with 0.549 d propagated uncertainty; this is association support with appreciable orbital uncertainty, not a precise phase coincidence. Next test: Optional precision characterization
R206	155916287	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Seven fitted repeats span S16,50,77. Sector-dependent depth variation and long-cadence integration limit intrinsic depth and duration precision. Next test: Optional precision characterization
R207	383603595	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Eleven fitted repeated eclipses span S8,62,90. Modest aperture/baseline depth variation limits precision; no missing bounded check. Next test: Optional precision characterization

Key	TIC	Population	Verdict	Added	Evidence / limitation
R208	436724968	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; QLP free-trapezoid depths show modest parity differences, especially in S71; local baseline and aperture dependence are material. Independent S43 3x3 cutout depths are 35.34-37.19 ppt for eight events including consecutive odd/even cycles, without a stable unequal alternating pattern. Gaia P=2.732167 d matches the measured event spacing. These data do not establish two distinct alternating eclipse kinds or require doubling; report the fitted period as the adopted orbit with this depth-model limitation, and the epoch as an observed reference eclipse. Capable neighbours at 35.19 and 35.89 arcsec are displaced from the compact target deficit. Next test: Optional precision characterization
R209	37948318	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; The historical absolute target-to-deficit offset is 15.1 arcsec (0.719 pixel), while light-to-deficit separation is 3.51 arcsec; these are distinct comparisons. The capable G13.73 neighbour is 53.65 arcsec away and opposite the deficit displacement. Independent per-event pixel checks find target-pixel losses 9.03-10.59 electrons/s against 0.185-0.437 at that neighbour in cycles317,319,321. This supports the target over the capable neighbour at TESS resolution, with an uncalibrated absolute-centroid systematic and residual crowding uncertainty. Historical cycles316/318 have common-mode excursions and are not individually decisive. QLP cycle93 has singular timing covariance and is excluded from timing. Next test: Optional precision characterization
R210	373034189	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Four raw fitted eclipses in S10,38,90 support the period. The Gaia conjunction uncertainty is substantial (0.3105 d); local model widths and depths are descriptive. Next test: Optional precision characterization
R211	300969635	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Deep repeated eclipses persist in S1,37,98. Aperture depth changes from about210 to248 ppt; pooled parity depths confound sector coverage and do not establish stable unequal eclipse kinds. Gaia period matches the measured spacing. Intrinsic depth requires dilution modeling. Next test: Optional precision characterization
R212	317800184	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Repeated eclipses span S6,87,98 with sector-dependent depths around45-62 ppt. This limits precision but does not erase the localized repeat pattern. Next test: Optional precision characterization
R213	219033622	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Three constrained same-kind events in S15 and S23 determine the ephemeris; S23 cycle32 is retained as a raw repeat but excluded from timing because sparse sampling leaves its egress poorly constrained. S76 did not supply a usable selected event window and is not a clean non-detection. The independent S16 cutout contains four repeating deep target-centered deficits. The timing baseline is shorter than the available sector list suggests. Next test: Optional precision characterization
R214	312725755	G 11.5-12 spectroscopic-orbit population	CONFIRMED	true	Report; Five fitted repeating eclipses span S19,45,73. The measured widths are long but compatible with approximate Gaia/TIC geometry; intrinsic depths and stellar parameters need detailed modeling. Next test: Optional precision characterization
P025	382042754	G 11.5-12; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim. Next test: Optional precision characterization; no missing bounded admission check
P026	389128842	G 11.5-12; Gaia spectroscopic orbit 20-60 d	CONFIRMED	true	Report; Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim. Next test: Optional precision characterization; no missing bounded admission check

Included probable systems (19)

Key	TIC	Population	Verdict	Added	Evidence / limitation
R003	298645930	G 10-11.5 spectroscopic-orbit population	PROBABLE	true	Report; Repeated eclipses secure; host attribution/check unresolved. See dated tier reconciliation and unchanged historical report.
R048	279166593	G 10-11.5 spectroscopic-orbit population	PROBABLE	true	Report; Repeated eclipses secure; host attribution/check unresolved. See dated tier reconciliation and unchanged historical report.
R058	236531158	G 10-11.5 spectroscopic-orbit population	PROBABLE	true	Report; Repeated eclipses secure; host attribution/check unresolved. See dated tier reconciliation and unchanged historical report.
R065	250742811	G 10-11.5 spectroscopic-orbit population	PROBABLE	true	Report; Repeated eclipses secure; host attribution/check unresolved. See dated tier reconciliation and unchanged historical report.
R067	351541670	G 10-11.5 spectroscopic-orbit population	PROBABLE	true	Report; Repeated eclipses secure; host attribution/check unresolved. See dated tier reconciliation and unchanged historical report.
R098	186100980	G 10-11.5 spectroscopic-orbit population	PROBABLE	true	Report; Repeated eclipses secure; host attribution/check unresolved. See dated tier reconciliation and unchanged historical report.

Key	TIC	Population	Verdict	Added	Evidence / limitation
R107	39069184	G 10-11.5 spectroscopic-orbit population	PROBABLE	true	Report; Repeated eclipses secure; host attribution/check unresolved. See dated tier reconciliation and unchanged historical report.
R136	101779765	G 10-11.5 spectroscopic-orbit population	PROBABLE	true	Report; Repeated eclipses secure; host attribution/check unresolved. See dated tier reconciliation and unchanged historical report.
R153	47719859	G 10-11.5 spectroscopic-orbit population	PROBABLE	true	Report; Refit the Gaia SB2 radial velocities jointly with the long-baseline eclipse ephemeris to resolve the orbital association.
C23	465263741	fainter SB1 period-fold population	PROBABLE	true	Report; See target report for deciding limitation and next check.
C29	285942031	fainter SB1 period-fold population	PROBABLE	true	Report; Reassess Gaia orbital uncertainties/correlations or obtain an updated radial-velocity orbit to resolve the eclipse-versus-Gaia timing tension.
C31	281393563	fainter SB1 period-fold population	PROBABLE	true	Report; Obtain higher resolution time-series imaging or a validated multi-source PRF difference fit to identify the eclipsing host.
C33	126363604	fainter SB1 period-fold population	PROBABLE	true	Report; Validate the repeated difference-image astrometry with a multi-source PRF or higher resolution event photometry; refine Gaia conjunction uncertainty if needed.
C44	201317466	fainter SB1 period-fold population	PROBABLE	true	Report; Refit the RV orbit or evaluate a nonlinear orbital posterior against the observed eclipse timing to resolve Gaia association.
C50	443106429	fainter SB1 period-fold population	PROBABLE	true	Report; Resolve target versusTIC 443106423 with higher spatial resolution eclipse photometry.
C67	9396327	fainter SB1 period-fold population	PROBABLE	true	Report; Refine Gaia conjunction predictions with orbital posterior and measure a second clean sector or obtain new RVs to test the orbit association.
R179	253126207	G 11.5-12 spectroscopic-orbit population	PROBABLE	true	Report; Attribution remains unresolved: seven capable neighbours include three within one TESS pixel (15.16, 17.15 and 19.43 arcsec); the historical PSF fractional residual is 0.657. A known Gaia eclipsing binary at 24.77 arcsec with P=0.312563 d is a different source, and its short period does not identify the 17.44 d signal. Real repeating eclipses pass the other checks; source localization is the failed scientific check. Resolved eclipse imaging or defensible multi-source PRF modeling is required. Next test: Resolved eclipse photometry / multi-source PRF localization
R215	167178300	12-14.8 d underreviewed Gaia period-fold pilot	PROBABLE	true	Report; Capable Gaia neighbours remain at 58.583arcsec, 41.62arcsec, 12.984arcsec, 24.468arcsec, 18.174arcsec, 59.428arcsec. Deficit centroid favors target, but close blend is unresolved without a PRF/deblend uncertainty test. Next test: Resolved eclipse photometry or multi-source PRF localization
R216	325573958	12-14.8 d underreviewed Gaia period-fold pilot	PROBABLE	true	Report; Capable Gaia neighbours remain at 57.77arcsec, 7.958arcsec, 20.628arcsec. Deficit centroid favors target, but close blend is unresolved without a PRF/deblend uncertainty test. Next test: Resolved eclipse photometry or multi-source PRF localization

Known recoveries and controls (8)

Key	TIC	Population	Verdict	Added	Evidence / limitation
R102	432482170	G 10-11.5 spectroscopic-orbit population	CONFIRMED	false	Report; See target report for limitations.
R163	407991939	G 10-11.5 spectroscopic-orbit population	CONFIRMED	false	Report; Joint RV/eclipse modeling and longer-baseline eclipse timing; resolve remaining alias and geometry uncertainty.
R165	407220461	G 10-11.5 spectroscopic-orbit population	CONFIRMED	false	Report; Higher-resolution eclipse photometry or calibrated PRF fitting to resolve the close capable neighbour; joint RV/eclipse timing fit.
C01	195865529	fainter SB1 period-fold population	CONFIRMED	false	Report; Known ASAS-SN EA at 0.16 arcsec (P 3.8310 d); ATLAS DBH P/2. Recovery excluded from new count.
C05	70665829	fainter SB1 period-fold population	CONFIRMED	false	Report; Already objects.csv A4; blind control independently recovered. Archive Gaia-VEB/varisum/SB9 controls failed on later mirror timeout, predecessor empty cones only. No addition.
C18	300903537	fainter SB1 period-fold population	CONFIRMED	false	Report; See target report for deciding limitation and next check.
C58	183994341	fainter SB1 period-fold population	CONFIRMED	false	Report; Resolve ASAS-SN period alias and obtain RV/orbital posterior or a joint timing/geometry analysis to settle the Gaia association.
P024	155303130	G 11.5-12; Gaia spectroscopic orbit 20-60 d	CONFIRMED	false	Report; Pilot characterization; period errors are scatter-only. No exhaustive literature priority claim. Next test: Optional precision characterization; no missing bounded admission check

Plausible signals (14)

Key	TIC	Population	Verdict	Added	Evidence / limitation
R064	1989681813	G 10-11.5 spectroscopic-orbit population	PLAUSIBLE	false	Report; Obtain spatially resolved time-series photometry across a later narrow deficit and both Gaia conjunctions; use empirical calibrated TESS PRFs in a second sector to distinguish astrophysical deficit from nonstationary spatial contamination.
C16	319185391	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Fold-only ~1 ppt dip, individual events ~1 sigma, dip not pixel-tested, no second pipeline. Report prose says probable-level but batch summary says PLAUSIBLE at best; adopt batch summary outside count. Gaia 434-day astrometric orbit suggests triple.
C49	71866523	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Require a clean repeated negative event before EB classification.
C54	92452368	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Verify a clean recurrent event in independently reduced data.
C55	94300366	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Obtain detrending resistant multiseason photometry to establish stable discrete repeats.
C57	402940895	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Re-extract clean sectors with robust background/pointing treatment before asserting eclipses.
C59	166781582	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.
C60	230270717	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.
C61	462845564	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.
C63	350661747	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.
C64	424565669	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Obtain or inspect another clean sector and require repeated discrete ingress/egress before fitting a binary period.
C68	275022619	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.
C72	257575960	fainter SB1 period-fold population	PLAUSIBLE	false	Report; Obtain a well-covered predicted candidate event in a distinct reduction and establish a second coherent detached eclipse before claiming an eclipse period or host.
P020	8853239	G 11-11.5; Gaia spectroscopic orbit 20-60 d	PLAUSIBLE	false	Report; Single secure event; Gaia-only period; KELT null is coverage-limited and not a refutation. Tentative B is not an ephemeris. Next test: Measure a second A eclipse to test the cycle period; a B detection alone would still need repeat timing.

Non-EB, neighbouring-source and systematic signals (21)

Key	TIC	Population	Verdict	Added	Evidence / limitation
C13	372683900	fainter SB1 period-fold population	ELLIPSOIDAL	false	Report; Smooth double wave, no narrow eclipse; unresolved 1.8 arcsec neighbour; Gaia eccentricity conflicts at 3.3 sigma.
C14	13106408	fainter SB1 period-fold population	ELLIPSOIDAL	false	Report; Smooth double wave; nearby 364-day SR unrelated.
C15	372683622	fainter SB1 period-fold population	ELLIPSOIDAL	false	Report; Double wave plus variable spots; no eclipse.
C17	62328710	fainter SB1 period-fold population	NEIGHBOUR_SIGNAL	false	Report; Delta Scuti-like signal localized to Gaia DR3 2165710355333626240 at 44 arcsec and independently seen there by ZTF; target flat at Gaia period; suspect SB1 orbit is interpretation.
C27	10790605	fainter SB1 period-fold population	REFUTED	false	Report; See target report for deciding limitation and next check.
C28	376082142	fainter SB1 period-fold population	REFUTED	false	Report; Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.
C34	251357000	fainter SB1 period-fold population	REFUTED	false	Report; Search higher precision data for a discrete narrow eclipse only if this smooth variable remains of interest.
C42	12320469	fainter SB1 period-fold population	REFUTED	false	Report; Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.
C43	86908196	fainter SB1 period-fold population	REFUTED	false	Report; Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.

Key	TIC	Population	Verdict	Added	Evidence / limitation
C46	411647453	fainter SB1 period-fold population	REFUTED	false	Report; Higher resolution event photometry can establish the contaminant host and measure its orbital period precisely.
C47	450217907	fainter SB1 period-fold population	REFUTED	false	Report; Higher precision light curves could test for a narrow residual eclipse if further work on this smooth variable is desired.
C48	60443793	fainter SB1 period-fold population	REFUTED	false	Report; Review TIC 60443695 separately if desired.
C51	311206604	fainter SB1 period-fold population	REFUTED	false	Report; No further eclipse follow-up required unless new discrete events appear.
C52	428769301	fainter SB1 period-fold population	REFUTED	false	Report; No further eclipse follow-up required unless new discrete events appear.
C53	436947190	fainter SB1 period-fold population	REFUTED	false	Report; No further eclipse follow-up required unless new discrete events appear.
C62	431762229	fainter SB1 period-fold population	REFUTED	false	Report; Higher resolution eclipse photometry can verify the contaminating host.
C65	222498305	fainter SB1 period-fold population	REFUTED	false	Report; Higher resolution eclipse photometry can verify the contaminating host.
C66	90542579	fainter SB1 period-fold population	REFUTED	false	Report; If revisiting, use clean cadences away from downlink boundaries with a validated background and aperture.
C69	297541839	fainter SB1 period-fold population	REFUTED	false	Report; No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.
C70	396248369	fainter SB1 period-fold population	REFUTED	false	Report; No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.
C71	277479128	fainter SB1 period-fold population	REFUTED	false	Report; No missing deciding gate for this candidate. Reopen only if new data supply distinct, repeated detached events.